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United States Patent Application Publication

20250283076

Kind Code

A1

Publication Date

September 11, 2025

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TREATMENT OF CARDIOVASCULAR DISEASE

Abstract

This disclosure relates to a nucleic acid comprising a double stranded RNA molecule comprising sense and antisense strands and further comprising a single stranded DNA molecule covalently linked to at least the 5' end of either the sense or antisense RNA part of the molecule wherein the double stranded inhibitory RNA targets genes associated with cardiovascular disease in the treatment hypercholesterolemia and diseases associated with hypercholesterolemia such as cardiovascular disease.

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Appl. No.: 18/691775

Filed (or PCT Filed): September 13, 2022

PCT No.: PCT/EP2022/075355

Foreign Application Priority Data

GB	2113104.0	Sep. 14, 2021
GB	2207239.1	May. 18, 2022

Publication Classification

Int. Cl.: C12N15/113 (20100101)

U.S. Cl.:

CPC **C12N15/113** (20130101); **C12N15/1137** (20130101); C12N2310/11 (20130101);
C12N2310/14 (20130101); C12N2310/312 (20130101); C12N2310/315 (20130101);
C12N2310/335 (20130101); C12N2310/3519 (20130101); C12N2310/531 (20130101)

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This is the U.S. National Stage of International Application No. PCT/EP2022/075355, filed Sep. 13, 2022, which was published in English under PCT Article 21(2), which claims the benefit of GB Application No. 2207239.1 filed May 18, 2022 and GB Application No. 2113104.0, filed Sep. 14, 2021.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates to a nucleic acid comprising a double stranded RNA molecule comprising sense and antisense strands and further comprising a single stranded DNA molecule covalently linked to at least 5' end of either the sense or antisense RNA part of the molecule wherein the double stranded inhibitory RNA targets of cardiovascular disease genes; pharmaceutical compositions comprising said nucleic acid molecule and methods for the treatment of diseases associated with increased levels of expression of cardiovascular disease genes, for example hypercholesterolemia.

INCORPORATION OF ELECTRONIC SEQUENCE LISTING

[0003] The electronic sequence listing, submitted herewith as an XML file named 4860P-US.xml (772,353 bytes), created on Sep. 26, 2024, is herein incorporated by reference in its entirety.

BACKGROUND TO THE DISCLOSURE

[0004] Cardiovascular disease associated with hypercholesterolemia, for example ischaemic cardiovascular disease, is a common condition and results in heart disease and a high incidence of death and morbidity and can be a consequence of poor diet, obesity, or an inherited dysfunctional gene. For example, high levels of lipoprotein (a) and other lipoproteins, is associated with atherosclerosis. Cholesterol is essential for membrane biogenesis in animal cells. The lack of water solubility means that cholesterol is transported around the body in association with lipoproteins. Apolipoproteins form together with phospholipids, cholesterol and lipids which facilitate the transport of lipids such as cholesterol, through the bloodstream to the different parts of the body. Lipoproteins are classified according to size and can form HDL (High-density lipoprotein), LDL (Low-density lipoprotein), IDL (intermediate-density lipoprotein), VLDL (very low-density lipoprotein) and ULDL (ultra-low-density lipoprotein) lipoproteins.

[0005] Lipoproteins change composition throughout their circulation comprising different ratios of apolipoproteins A (ApoA), B (ApoB), C (ApoC), D(ApoD) or E (ApoE), triglycerides, cholesterol and phospholipids. For example, ApoB is the main apolipoprotein of ULDL and LDL and has two isoforms apoB-48 and apoB-100. Both ApoB isoforms are encoded by one single gene and wherein the shorter ApoB-48 gene is produced after RNA editing of the ApoB-100 transcript at residue 2180 resulting in the creation of a stop codon. ApoB-100 is the main structural protein of LDL and serves as a ligand for a cell receptor which allows transport of, for example, cholesterol into a cell.

[0006] Familial hypercholesterolemia is an orphan disease and results from elevated levels of LDL cholesterol (LDL-C) in the blood. The disease is an autosomal dominant disorder with both the heterozygous (350-550 mg/dL LDL-C) and homozygous (650-1000 mg/dL LDL-C) states resulting in elevated LDL-C. The heterozygous form of familial hypercholesterolemia is around 1:500 of the population. The homozygous state is much rarer and is approximately 1:1,000,000. The normal levels of LDL-C are in the region 130 mg/dL.

[0007] Hypercholesterolemia is particularly acute in paediatric patients which if not diagnosed early can result in accelerated coronary heart disease and premature death. If diagnosed and treated early the child can have a normal life expectancy. In adults, high LDL-C, either because of mutation or other factors, is directly associated with increased risk of atherosclerosis which can lead to coronary artery disease, stroke, or kidney disease. Lowering levels of LDL-C is known to reduce the risk of atherosclerosis and associated conditions. LDL-C levels can be lowered initially by administration of statins which block the de novo synthesis of cholesterol by inhibiting the HMG-CoA reductase. Some subjects can benefit from combination therapy which combines a statin with other therapeutic agents such as ezetimibe, colestipol or nicotinic acid. However, expression and synthesis of HMG-CoA reductase adapts in response to the statin inhibition and increases over time, thus the beneficial effects are only temporary or limited after statin resistance is established.

[0008] There is therefore a desire to identify alternative therapies that can be used alone or in combination with existing therapeutic approaches to control cardiovascular disease because of elevated LDL-C.

[0009] A technique to specifically ablate gene function is through the introduction of double stranded inhibitory RNA, also referred to as small inhibitory or interfering RNA (siRNA), into a cell which results in the destruction of mRNA complementary to the sequence included in the siRNA molecule. The siRNA molecule comprises two complementary strands of RNA (a sense strand and an antisense strand) annealed to each other to form a double stranded RNA molecule. The siRNA molecule is typically, but not exclusively, derived from exons of the gene which is to be ablated. Many organisms respond to the presence of double stranded RNA by activating a cascade that leads to the formation of siRNA. The presence of double stranded RNA activates a protein complex comprising RNase III which processes the double stranded RNA into smaller fragments (siRNAs, approximately 21-29 nucleotides in length) which become part of a ribonucleoprotein complex. The siRNA acts as a guide for the RNase complex to cleave mRNA complementary to the antisense strand of the siRNA thereby resulting in destruction of the mRNA.

[0010] The inhibition of expression of lipoprotein (a) is known and the use of inhibitory RNA to silence expression of lipoprotein (a) is also known. For example, WO2019/092283 discloses the identification of specific siRNA sequences that target knock down of mRNA encoding lipoprotein (a) and their use in the treatment of cardiovascular diseases linked to elevated lipoprotein (a) expression such as coronary heart disease, aortic stenosis or stroke. Similarly, U.S. Pat. No. 9,932,586 discloses specific siRNA sequences that target lipoprotein (a) expression and their use in the treatment of cardiovascular diseases linked to elevated lipoprotein (a) expression such as Buerger's disease, coronary heart disease, renal artery stenosis, hyperapobetalipoproteinemia, cerebrovascular atherosclerosis, cerebrovascular disease, and venous thrombosis.

[0011] Over expression of APOC III is associated with atherosclerosis and type 2 diabetes. For example, WO2003/020765 discloses a vaccination approach to the control of atherosclerosis using immunogens derived from ApoCIII polypeptide and its use in controlling atherosclerotic plaques in coronary and cerebrovascular disease. A similar vaccination approach is disclosed 5 in WO2004/080375 and WO2001/064008. In WO2014/205449 and WO2014/179626 is disclosed the use of antisense oligonucleotides to improve insulin sensitivity and treat type II diabetes by targeting APOCIII expression.

[0012] Furthermore, WO2007/136989 and WO2005/019418 each disclose the use of antisense compounds directed to DGAT to regulate expression of DGAT2 and treat conditions that would benefit from reduction in DGAT2 expression in relation to conditions that would benefit from reduction in serum triglyceride levels such as hypercholesterolemia, cardiovascular disease, type II diabetes and metabolic syndrome. WO2018/093966 discloses the use of RNA silencing 10 directed to DGAT2 and diglyceride acyltransferase 1(DGAT1) to treat obesity and obesity associated diseases such as hypercholesterolemia, cardiovascular disease, type II diabetes and metabolic

syndrome. Similarly, WO2005/044981 discloses the use of siRNA to target DGAT2 amongst many other gene targets and their use in the treatment of diseases that would benefit from triglyceride regulation.

[0013] This disclosure relates to a nucleic acid molecule comprising a double stranded inhibitory RNA that is modified by the inclusion of a short DNA part linked to at least the 5' end of either the sense or antisense inhibitory RNA and which forms a hairpin structure. The double stranded inhibitory RNA uses solely or predominantly natural nucleotides and does not require modified nucleotides or sugars that prior art double stranded RNA molecules typically utilise to improve pharmacodynamics and pharmacokinetics. The disclosed double stranded inhibitory RNAs have activity in silencing cardiovascular gene targets with potentially fewer side effects.

STATEMENTS OF THE INVENTION

[0014] According to an aspect of the invention there is provided a nucleic acid molecule comprising [0015] a first part that comprises a double stranded inhibitory ribonucleic acid (RNA) molecule comprising a sense strand and an antisense strand; and [0016] a second part that comprises a single stranded deoxyribonucleic acid (DNA) molecule, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule or wherein the 3' end of the single stranded DNA molecule is covalently linked to the 5' of the antisense strand of the double stranded inhibitory RNA molecule, characterized in that the double stranded inhibitory RNA comprises a sense nucleotide sequence that encodes a part of a cardiovascular gene target associated with cardiovascular disease and wherein said single stranded DNA molecule comprises a nucleotide sequence that is adapted over at least part of its length to anneal by complementary base pairing to a part of said single stranded DNA to form a double stranded DNA structure wherein said double stranded inhibitory RNA consists of natural nucleotides.

[0017] According to an aspect of the invention there is provided a nucleic acid molecule comprising [0018] a first part that comprises a double stranded inhibitory ribonucleic acid (RNA) molecule comprising a sense strand and an antisense strand; and [0019] a second part that comprises a single stranded deoxyribonucleic acid (DNA) molecule, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule or wherein the 3' end of the single stranded DNA molecule is covalently linked to the 5' of the antisense strand of the double stranded inhibitory RNA molecule, characterized in that the double stranded inhibitory RNA comprises a sense nucleotide sequence that encodes a part of a cardiovascular gene target associated with cardiovascular disease, or a polymorphic sequence variant thereof, and wherein said single stranded DNA molecule comprises a nucleotide sequence that is adapted over at least part of its length to anneal by complementary base pairing to a part of said single stranded DNA to form a double stranded DNA structure comprising a stem and a loop domain, characterized in that said nucleic acid molecule comprises N-acetylgalactosamine and said double stranded inhibitory RNA consists of natural nucleotides.

[0020] A "polymorphic sequence variant" is a gene sequence that varies by one, two, three or more nucleotides.

[0021] In a preferred embodiment of the invention wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule.

[0022] In a preferred embodiment of the invention wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

[0023] In a preferred embodiment of the invention single stranded DNA molecule is covalently linked to the 5' end of said sense strand and the 5' end of said antisense strand.

[0024] In an alternative embodiment of the invention said single stranded DNA molecule is covalently linked to the 5' end of said sense strand, the 3' end of said sense strand.

[0025] In a preferred embodiment of the invention said loop portion comprises a region comprising the nucleotide sequence GNA or GNNA, wherein each N independently represents guanine (G), thymidine (T), adenine (A), or cytosine (C).

[0026] In a preferred embodiment of the invention said loop domain comprises G and C nucleotide bases.

[0027] In an alternative embodiment of the invention said loop domain comprises the nucleotide sequence GCGAAGC.

[0028] In a preferred embodiment of the invention said single stranded DNA molecule comprises the nucleotide sequence 5TCACCTCATCCCGCGAAGC 3' (SEQ ID NO 387 and 251).

[0029] In a preferred embodiment of the invention said single stranded DNA molecule comprises the nucleotide sequence 5' CGAAGCGCCCTACTCCACT 3'. (SEQ ID NO 130)

[0030] In a preferred embodiment of the invention said single stranded DNA molecule comprises the nucleotide sequence 5' GCGAAGCCCCTACTCCACT 3' (SEQ ID NO 400).

[0031] The inhibitory RNA molecules comprise or consist of natural nucleotide bases that do not require chemical modification. Moreover, in some embodiments of the invention, wherein the crook DNA molecule is linked to the 3' end of the sense strand of said double stranded inhibitory RNA, the antisense strand is optionally provided with at least a two-nucleotide base overhang sequence. The two-nucleotide overhang sequence can correspond to nucleotides encoded by the target or are non-encoding. The two-nucleotide overhang can be two nucleotides of any sequence and in any order, for example UU, AA, UA. AU. GG, CC, GC, CG, UG, GU, UC, CU, and TT.

[0032] In a preferred embodiment of the invention said inhibitory RNA molecule comprises a two-nucleotide overhang comprising or consisting of deoxythymidine dinucleotide (dTdT).

[0033] In a preferred embodiment of the invention said dTdT overhang is positioned at the 5' end of said antisense strand.

[0034] In an alternative preferred embodiment of the invention said dTdT overhang is positioned at the 3' end of said antisense strand.

[0035] In a preferred embodiment of the invention said dTdT overhang is positioned at the 5' end of said sense strand.

[0036] In an alternative preferred embodiment of the invention said dTdT overhang is positioned at the 3' end of said sense strand.

[0037] In a preferred embodiment of the invention said sense and/or said antisense strands comprises internucleotide phosphorothioate linkages.

[0038] In a preferred embodiment of the invention said sense strand comprises internucleotide phosphorothioate linkages.

[0039] Preferably, the 5' and/or 3' terminal two nucleotides of said sense strand comprises two internucleotide phosphorothioate linkage.

[0040] In a preferred embodiment of the invention said antisense strand comprises internucleotide phosphorothioate linkages.

[0041] Preferably, the 5' and/or 3' terminal two nucleotides of said antisense strand comprises two internucleotide phosphorothioate linkages.

[0042] In an alternative preferred embodiment of the invention said single stranded DNA molecule comprises one or more internucleotide phosphorothioate linkages.

[0043] In a preferred embodiment of the invention said nucleic acid molecule comprises a vinylphosphonate modification.

[0044] In a preferred embodiment of the invention said vinylphosphonate modification is to the 5' terminal phosphate of said sense RNA strand.

[0045] In a preferred embodiment of the invention said vinylphosphonate modification is to the 5' terminal phosphate of said antisense RNA strand.

[0046] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule is between 10 and 40 nucleotides in length.

[0047] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule is between 17 and 29 nucleotides in length.

[0048] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule is 19 to 21 nucleotides in length. Preferably, 19 nucleotides in length.

[0049] In a preferred embodiment of the invention said cardiovascular gene target is Human Lipoprotein (a).

[0050] In an alternative embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence selected from the group consisting of: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33 or 34.

[0051] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 41 and a sense nucleotide sequence comprising SEQ ID NO: 49, wherein said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule.

[0052] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 4 and a sense nucleotide sequence comprising SEQ ID NO: 44, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

[0053] In a preferred embodiment of the invention said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 5 and a sense nucleotide sequence comprising SEQ ID NO: 46, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

[0054] In an alternative preferred embodiment of the invention said cardiovascular gene target is Human Apolipoprotein C III (Apo C III).

[0055] Preferably, said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78 and 79.

[0056] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249 and 250.

[0057] Preferably said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 50, 51, 52, 53, 54, 55, 56, 57, 58, 80, 81, 82, 83, 84, 85, 86, 87, 88 and 89.

[0058] In an alternative preferred embodiment of the invention said cardiovascular gene target is Human diglyceride acyltransferase 2 (DGAT2).

[0059] Preferably, said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118 and 119.

[0060] Preferably, said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169 and 170.

[0061] Preferably said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 120, 121, 122, 123, 124, 125, 126, 127, 128 and 129.

[0062] In a preferred embodiment of the invention said cardiovascular gene target is Human PCSK9.

[0063] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 171,

172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 189 and 190.

[0064] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209 and 210.

[0065] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 255, 256, 257, 258, 259, 260, 261, 262, 263 and 264.

[0066] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 265, 266, 267, 268, 269, 270, 271, 272, 273 and 274.

[0067] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 275, 276, 277, 278, 279, 280, 281, 282, 283, 284, 285, 286, 287, 288, 289, 290, 292, 292, 293, 294, 295, 296, 297, 298, 299, 300, 301, 302, 303, 304, 305, 306, 307, 308, 309, 310, 311, 312, 313, 314, 315, 316, 317, 318, 319, 320, 321, 322, 323, 324, 325, 326, 327, 328, 329 and 330.

[0068] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 331, 332, 333, 334, 335, 336, 337, 338, 339, 340, 341, 342, 343, 344, 345, 346, 347, 348, 349, 350, 351, 352, 353, 354, 355, 356, 357, 358, 359, 360, 361, 362, 363, 364, 365, 366, 367, 368, 369, 370, 371, 372, 373, 374, 375, 376, 377, 378, 379, 380, 381, 382, 383, 384, 285 and 386.

[0069] In a preferred embodiment of the invention said cardiovascular gene target is Human Apolipoprotein B.

[0070] In a preferred embodiment of the invention said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 499, 500, 453, 502, 503, 457, 505, 506, 462, 508, 509, 467, 511, 512, 472, 514, 515, 477, 517 518, 482, 520, 521, 487, 523, 524 and 492.

[0071] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising a nucleotide sequence selected from the group consisting of: 450, 501, 455, 504, 460, 507, 465, 510, 470, 513, 475, 516, 480, 519, 485, 522, 490 and 525.

[0072] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 1.

[0073] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 2.

[0074] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 3.

[0075] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 4.

[0076] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 5.

[0077] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 8.

[0078] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 10.

[0079] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 14.

[0080] In a preferred embodiment of the invention said nucleic acid molecule comprises a RNA strand comprising or consisting of a nucleotide sequence, or polymorphic sequence variant, set forth in table 15.

[0081] In a preferred embodiment of the invention said nucleic acid molecule comprises or consists of between 19 and 21 contiguous nucleotides of the nucleotide sequence set forth in SEQ ID NO:388.

[0082] In a preferred embodiment of the invention said nucleic acid molecule is covalently linked to N-acetylgalactosamine.

[0083] In a further embodiment of the invention N-acetylgalactosamine is linked to either the antisense part of said inhibitory RNA or the sense part of said inhibitory RNA.

[0084] Preferably, N-acetylgalactosamine is linked to the 5' terminus of said sense RNA. In an alternative embodiment of the invention N-acetylgalactosamine is linked to the 3' terminus of said sense RNA.

[0085] In an alternative preferred embodiment of the invention said N-acetylgalactosamine is linked to the 3' terminus of said antisense RNA.

[0086] In a preferred embodiment of the invention N-acetylgalactosamine is monovalent.

[0087] In a preferred embodiment of the invention N-acetylgalactosamine is divalent.

[0088] In an alternative embodiment of the invention N-acetylgalactosamine is trivalent.

[0089] In a preferred embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:

##STR00001##

[0090] In an alternative embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:

##STR00002##

[0091] In an alternative embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:

##STR00003##

[0092] In an alternative embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising the structure:

##STR00004##

[0093] In an alternative preferred embodiment of the invention said nucleic acid molecule is covalently linked to a molecule comprising N-acetylgalactosamine 4-sulfate.

[0094] According to a further aspect of the invention there is provided a pharmaceutical composition comprising at least one nucleic acid molecule according to the invention.

[0095] In a preferred embodiment of the invention said composition further includes a pharmaceutical carrier and/or excipient.

[0096] When administered the compositions of the present invention are administered in pharmaceutically acceptable preparations. Such preparations may routinely contain pharmaceutically acceptable concentrations of salt, buffering agents, preservatives, compatible carriers and optionally other therapeutic agents, such as cholesterol lowering agents, which can be administered separately from the nucleic acid molecule according to the invention or in a combined preparation if a combination is compatible.

[0097] The combination of a nucleic acid according to the invention and the other, different therapeutic agent is administered as simultaneous, sequential or temporally separate dosages.

[0098] The therapeutics of the invention can be administered by any conventional route, including injection or by gradual infusion over time. The administration may, for example, be oral, intravenous, intraperitoneal, intramuscular, intracavity, subcutaneous, transdermal or

transepithelial.

[0099] The compositions of the invention are administered in effective amounts. An “effective amount” is that amount of a composition that alone, or together with further doses, produces the desired response. In the case of treating a disease, such as cardiovascular disease, the desired response is inhibiting or reversing the progression of the disease. This may involve only slowing the progression of the disease temporarily, although more preferably, it involves halting the progression of the disease permanently. This can be monitored by routine methods.

[0100] Such amounts will depend, of course, on the particular condition being treated, the severity of the condition, the individual patient parameters including age, physical condition, size and weight, the duration of the treatment, the nature of concurrent therapy (if any), the specific route of administration and like factors within the knowledge and expertise of the health practitioner. These factors are well known to those of ordinary skill in the art and can be addressed with no more than routine experimentation. It is generally preferred that a maximum dose of the individual components or combinations thereof be used, that is, the highest safe dose according to sound medical judgment. It will be understood by those of ordinary skill in the art, however, that a patient may insist upon a lower dose or tolerable dose for medical reasons, psychological reasons or for virtually any other reasons.

[0101] The pharmaceutical compositions used in the foregoing methods preferably are sterile and contain an effective amount of a nucleic acid molecule according to the invention for producing the desired response in a unit of weight or volume suitable for administration to a patient. The response can, for example, be measured by determining regression of cardiovascular disease and decrease of disease symptoms etc.

[0102] The doses of the nucleic acid molecule according to the invention administered to a subject can be chosen in accordance with different parameters, in particular in accordance with the mode of administration used and the state of the subject. Other factors include the desired period of treatment. If a response in a subject is insufficient at the initial doses applied, higher doses (or effectively higher doses by a different, more localized delivery route) may be employed to the extent that patient tolerance permits. It will be apparent that the method of detection of the nucleic acid according to the invention facilitates the determination of an appropriate dosage for a subject in need of treatment.

[0103] In general, doses of the nucleic acid molecules herein disclosed of between 1 nM-1 μ M generally will be formulated and administered according to standard procedures. Preferably doses can range from 1 nM-500 nM, 5 nM-200 nM, 10 nM-100 nM. Other protocols for the administration of compositions will be known to one of ordinary skill in the art, in which the dose amount, schedule of injections, sites of injections, mode of administration and the like vary from the foregoing. The administration of compositions to mammals other than humans, (e.g. for testing purposes or veterinary therapeutic purposes), is carried out under substantially the same conditions as described above. A subject, as used herein, is a mammal, preferably a human, and including a non-human primate or a transgenic mammal adapted for expression of human lipoprotein(a).

[0104] When administered, the pharmaceutical preparations of the invention are applied in pharmaceutically acceptable amounts and in pharmaceutically acceptable compositions. The term “pharmaceutically acceptable” means a non-toxic material that does not interfere with the effectiveness of the biological activity of the active ingredients. Such preparations may routinely contain salts, buffering agents, preservatives, compatible carriers, and optionally other therapeutic agents e.g., statins. When used in medicine, the salts should be pharmaceutically acceptable, but non-pharmaceutically acceptable salts may conveniently be used to prepare pharmaceutically acceptable salts thereof and are not excluded from the scope of the invention. Such pharmacologically and pharmaceutically acceptable salts include, but are not limited to, those prepared from the following acids: hydrochloric, hydrobromic, sulfuric, nitric, phosphoric, maleic, acetic, salicylic, citric, formic, malonic, succinic, and the like. Also, pharmaceutically acceptable

salts can be prepared as alkaline metal or alkaline earth salts, such as sodium, potassium, or calcium salts.

[0105] Compositions may be combined, if desired, with a pharmaceutically acceptable carrier. The term “pharmaceutically acceptable carrier” as used herein means one or more compatible solid or liquid fillers, diluents or encapsulating substances which are suitable for administration into a human. The term “pharmaceutically acceptable carrier” in this context denotes an organic or inorganic ingredient, natural or synthetic, with which the active ingredient is combined to facilitate, for example, solubility and/or stability. The components of the pharmaceutical compositions also are capable of being co-mingled with the molecules of the present invention, and with each other, in a manner such that there is no interaction which would substantially impair the desired pharmaceutical efficacy.

[0106] The pharmaceutical compositions may contain suitable buffering agents, including acetic acid in a salt; citric acid in a salt; boric acid in a salt; and phosphoric acid in a salt. The pharmaceutical compositions also may contain, optionally, suitable preservatives.

[0107] The pharmaceutical compositions may conveniently be presented in unit dosage form and may be prepared by any of the methods well-known in the art of pharmacy. All methods include the step of bringing the active agent into association with a carrier which constitutes one or more accessory ingredients. In general, the compositions are prepared by uniformly and intimately bringing the active compound into association with a liquid carrier, a finely divided solid carrier, or both, and then, if necessary, shaping the product. Compositions suitable for oral administration may be presented as discrete units, such as capsules, tablets, lozenges, each containing a predetermined amount of the active compound.

[0108] Compositions suitable for parenteral administration conveniently comprise a sterile aqueous or non-aqueous preparation of nucleic acid, which is preferably isotonic with the blood of the recipient. This preparation may be formulated according to known methods using suitable dispersing or wetting agents and suspending agents. The sterile injectable preparation also may be a sterile injectable solution or suspension in a non-toxic parenterally acceptable diluent or solvent, for example, as a solution in 1, 3-butane diol. Among the acceptable solvents that may be employed are water, Ringer's solution, and isotonic sodium chloride solution. In addition, sterile, fixed oils are conventionally employed as a solvent or suspending medium. For this purpose, any bland fixed oil may be employed including synthetic mono- or di-glycerides. In addition, fatty acids such as oleic acid may be used in the preparation of injectables. Carrier formulation suitable for oral, subcutaneous, intravenous, intramuscular, etc. administrations can be found in Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, PA.

[0109] In a further preferred embodiment of the invention said pharmaceutical composition comprises at least one further, different, therapeutic agent.

[0110] In a preferred embodiment of the invention said further therapeutic agent is a statin.

[0111] Statins are commonly used to control cholesterol levels in subjects that have elevated LDL-C. Statins are effective in preventing and treating those subjects that are susceptible and those that have cardiovascular disease. The typical dosage of a statin is in the region 5 to 80 mg but this is dependent on the statin and the desired level of reduction of LDL-C required for the subject suffering from high LDL-C. However, expression and synthesis of HMG-CoA reductase, the target for statins, adapts in response to statin administration thus the beneficial effects of statin therapy are only temporary or limited after statin resistance is established.

[0112] Preferably said statin is selected from the group consisting of atorvastatin, fluvastatin, lovastatin, pitvastatin, pravastatin, rosuvastatin and simvastatin.

[0113] In a preferred embodiment of the invention said further therapeutic agent is ezetimibe.

[0114] Optionally, ezetimibe is combined with at least one statin, for example simvastatin.

[0115] In an alternative preferred embodiment of the invention said further therapeutic agent is selected from the group consisting of fibrates, nicotinic acid, cholestyramine.

[0116] In a further alternative preferred embodiment of the invention said further therapeutic agent is a therapeutic antibody, for example, evolocumab, bococizumab or alirocumab.

[0117] According to a further aspect of the invention there is provided a nucleic acid molecule or a pharmaceutical composition according to the invention for use in the treatment or prevention of a subject that has or is predisposed to hypercholesterolemia or diseases associated with hypercholesterolemia.

[0118] In a preferred embodiment of the invention said subject is a paediatric subject.

[0119] A paediatric subject includes neonates (0-28 days old), infants (1-24 months old), young children (2-6 years old) and prepubescent [7-14 years old] children.

[0120] In an alternative preferred embodiment of the invention said subject is an adult subject.

[0121] In a preferred embodiment of the invention the hypercholesterolemia is familial hypercholesterolemia.

[0122] In a preferred embodiment of the invention familial hypercholesterolemia is associated with elevated levels of lipoprotein (a) expression.

[0123] In a preferred embodiment of the invention said subject is resistant to statin therapy.

[0124] In a preferred embodiment of the invention said disease associated with hypercholesterolemia is selected from the group consisting of: stroke prevention, hyperlipidaemia, cardiovascular disease, atherosclerosis, coronary heart disease, aortic stenosis, cerebrovascular disease, peripheral arterial disease, hypertension, metabolic syndrome, type II diabetes, non-alcoholic fatty acid liver disease, non-alcoholic steatohepatitis, Buerger's disease, renal artery stenosis, hyperapobetalipoproteinemia, cerebrovascular atherosclerosis, cerebrovascular disease and venous thrombosis.

[0125] According to a further aspect of the invention there is provided a method to treat a subject that has or is predisposed to hypercholesterolemia comprising administering an effective dose of a nucleic acid or a pharmaceutical composition according to the invention thereby treating or preventing hypercholesterolemia.

[0126] In a preferred method of the invention said subject is a paediatric subject.

[0127] In an alternative preferred method of the invention said subject is an adult subject.

[0128] In a preferred method of the invention the hypercholesterolemia is familial hypercholesterolemia.

[0129] In a preferred method of the invention familial hypercholesterolemia is associated with elevated levels of lipoprotein (a) expression.

[0130] In a preferred method of the invention said subject is resistant to statin therapy.

[0131] In a preferred method of the invention said disease associated with hypercholesterolemia is selected from the group consisting of: stroke prevention, hyperlipidaemia, cardiovascular disease, atherosclerosis, coronary heart disease, aortic stenosis, cerebrovascular disease, peripheral arterial disease, hypertension, metabolic syndrome, type II diabetes, non-alcoholic fatty acid liver disease, non-alcoholic steatohepatitis, Buerger's disease, renal artery stenosis, hyperapobetalipoproteinemia, cerebrovascular atherosclerosis, cerebrovascular disease and venous thrombosis.

[0132] Throughout the description and claims of this specification, the words “comprise” and “contain” and variations of the words, for example “comprising” and “comprises”, means “including but not limited to” and is not intended to (and does not) exclude other moieties, additives, components, integers or steps.

[0133] Throughout the description and claims of this specification, the singular encompasses the plural unless the context otherwise requires. In particular, where the indefinite article is used, the specification is to be understood as contemplating plurality as well as singularity, unless the context requires otherwise.

[0134] Features, integers, characteristics, compounds, chemical moieties or groups described in conjunction with an aspect, embodiment or example of the invention are to be understood to be applicable to any other aspect, embodiment or example described herein unless incompatible

therewith.

[0135] An embodiment of the invention will now be described by example only and with reference to the following figures.

Description

BRIEF SUMMARY OF THE DRAWINGS

[0136] FIG. 1 Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds. HepG2 cells were transfected with the following siRNAs after 30 mins or 2 hr incubation at 37 C in water, 10% FBS or 10% human serum: modified Inclisiran [white bar; SEQ ID NOs: 494 and 495], unmodified 'Inclisiran' with no Crook [grey bar; SEQ ID NOs: 389 and 390], unmodified Inclisiran with 3' SS Crook [hatched bar; SEQ ID NOs: 496 and 390], unmodified Inclisiran with 5' SS 'reversed hairpin' Crook [spotted bar; SEQ ID NOs: 497 and 390], or unmodified Inclisiran with 5' SS Crook [hatched bar; SEQ ID NOs: 498 and 390]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' treatment [black bar];

[0137] FIG. 2 Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds. HepG2 cells were transfected with the following siRNAs after a 2 hr incubation at 37 C in water, 10%, 20% or 50% FBS: modified Inclisiran [white bar; SEQ ID NOs: 494 and 495], unmodified 'Inclisiran' with no Crook [grey bar; SEQ ID NOs: 389 and 390], unmodified Inclisiran with 5' SS 'reversed hairpin' Crook [spotted bar; SEQ ID NOs: 497 and 390], or unmodified Inclisiran with 5' SS Crook [striped bar; SEQ ID NOs: 498 and 390]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' pre-treatment [black bar];

[0138] FIG. 3 Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds. HepG2 cells were transfected with the following siRNAs after a 4-hr incubation at 37 C in water, 10% FBS or 10% human serum: modified Inclisiran [white bar; SEQ ID NOs: 494 and 495], unmodified 'Inclisiran' with no Crook [grey bar; SEQ ID NOs: 389 and 390], unmodified Inclisiran with 5' SS 'reversed hairpin' Crook [spotted bar; SEQ ID NOs: 497 and 390], or unmodified Inclisiran with 5' SS Crook [striped bar; SEQ ID NOs: 498 and 390]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' pre-treatment [black bar];

[0139] FIG. 4 Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA (termed PC8-PC18) compounds. HepG2 cells were transfected with the following unmodified PC8-18 siRNAs after a 2-hr incubation at 37 C in water, 10% FBS or 10% human serum: siRNA35 with no Crook [white bar; SEQ ID NOs: 176 and 272], siRNA36 with no Crook but including dTdT overhangs on 3' SS & 3' AS [grey bar; SEQ ID NOs: 421 and 422], siRNA37 with Crook on 3' SS [spotted bar; SEQ ID NOs: 423 and 272], siRNA38 with Crook on 3' AS [vertical striped bar; SEQ ID NOs: 278 and 424], siRNA39 with Crook on 3' SS and dTdT overhang on 3' AS [hatched bar SEQ ID NOs: 423 and 422], siRNA41 with 5' SS 'reversed hairpin' Crook [horizontal stripe bar; SEQ ID NOs: 425 and 272], or siRNA42 with Crook on 5' SS and dTdT overhang on 3' AS [spots of black background bar; SEQ ID NOs: 425 and 422]. PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' pre-treatment [black bar];

[0140] FIG. 5A In vivo silencing of liver PCSK9 mRNA following administration of unmodified siRNA compounds (termed PC2-PC12) Groups of 5 mice for each treatment group were injected subcutaneously (SC) with either vehicle [black bar], compound A (no Crook; white bar; SEQ ID NOs: 172 and 252), compound G (Crook on 5' end of sense strand (SS); spotted bar; SEQ ID NOs: 428 and 252), or compound H (Crook on 3' end of SS; grey bar; SEQ ID NOs: 253 and 252). Each

compound was given at either 2 mg/kg or 10 mg/kg, and following sacrifice, levels of liver PCSK9 mRNA by RT-qPCR were measured at two time points (day 2 and day 7) and [0141] FIG. 5B Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds A, G and H, used in mouse in vivo study (FIG. 5A). HepG2 cells were transfected with siRNA compounds A, G, or H after 30 min or 2 hr incubation at 37 C in water, 10% FBS or 10% human serum: compound A (no Crook; white bar; SEQ ID NOs: 172 and 252), compound G (Crook on 5' end of sense strand (SS); spotted bar; SEQ ID NOs: 428 and 252), or compound H (Crook on 3' end of SS; grey bar; SEQ ID NOs: 253 and 252). PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' [black bar], and 'no serum' pre-treatment.

[0142] FIG. 5C Serum stability assays showing target PCSK9 mRNA levels in HepG2 cells following transfection of siRNA compounds A, G and H, used in mouse in vivo study (FIG. 5A). HepG2 cells were transfected with siRNA compounds A, G, or H after a 2 hr incubation at 37 C in water, 20% or 50% human serum: compound A (no Crook; white bar; SEQ ID NOs: 172 and 252), compound G (Crook on 5' end of sense strand (SS); spotted bar; SEQ ID NOs: 428 and 252), or compound H (Crook on 3' end of SS; grey bar; SEQ ID NOs: 253 and 252). PCSK9 mRNA levels were quantified by RT-qPCR analysis. Controls include 'no siRNA' [black bar], and 'no serum' pre-treatment.

MATERIALS AND METHODS

Hepg2 Reverse Transfection

[0143] Duplex siRNAs synthesized by Bio-Synthesis (Lewisville, TX) (Table 1), were resuspended in Nuclease-free water (Invitrogen™ AM9932) to generate a stock solution of 10 μM. For serum stability assay, stock siRNAs were incubated at 37° C. in vehicle (nuclease-free water), 10% fetal bovine serum (FBS) or in various concentrations (10%-80%) of human serum (HS) for 2 hours. After pre-incubation in serum or vehicle, siRNAs were transfected into HepG2 cells in a 384-well plate (Thermo Scientific™ 164688) at a concentration of 25 nM using 0.15 μL of Lipofectamine RNAiMAX (Invitrogen™ 13778075) per well. Transfected cells were incubated at 37° C. and 5% CO₂. Cells receiving no siRNA treatment were used as control.

Free-Uptake and Transfection in Primary Mouse Hepatocytes

[0144] Mouse hepatocytes (MSCP10, Lonza) were thawed and seeded in a 384-well plate (Thermo Scientific™ 164688) in Williams E medium (Gibco™ A1217601) supplemented with Primary Hepatocyte Thawing and Plating Supplements (Gibco™ CM3000). Cells were treated with siRNAs at 25 nM using 0.15 μL of Lipofectamine per well or with 100 nM of GalNAc-siRNAs for free-uptake.

Duplex RT-qPCR

[0145] Cells were processed for RT-qPCR read-out using the Cells-to-CT 1-step TaqMan Kit (Invitrogen™ A25603). Briefly, cells were washed with 50 μL ice-cold PBS and lysed in 20 μL Lysis solution containing DNase I. Lysis was stopped after 5 minutes by addition of 2 μL STOP Solution for 2 min. For the RT-qPCR analysis, 1 μL of lysate was dispensed per well into a 96-well PCR plate in a 20 μL RT-qPCR reaction volume. RT-qPCR was performed using the TaqMan® 1-Step qRT-PCR Mix from the Cells-to-CT 1-step TaqMan Kit, with TaqMan probes for GAPDH (VIC_PL, Assay Id Hs00266705_g1) and PCSK9 (FAM, Assay Id Hs00545399-m1) or ApoB (FAM, Assay Id Mm01545150_m1). RT-qPCR was performed using a QuantStudio 5 thermocycling instrument (Applied BioSystems). Relative quantification was determined using the $\Delta\Delta CT$ method, where GAPDH was used as internal control and expression changes normalized to the reference sample (no siRNA treatment).

In Vivo Mouse Study Animals

[0146] Male C57BL/6J mice (20-25 g) were group housed in the Saretius animal unit at the University of Reading, and maintained under a 12 h light/dark cycle, at 23° C. with humidity controlled according to Home Office regulations. Mice were given access to standard rodent chow

SDS rat expanded diet (RM3-E-FG) for the duration of the study.

Formulation of siRNA Compounds

[0147] Compound A, Compound G, and Compound H were each formulated in RNAase free PBS to concentrations of 0.4 and 2 mg/mL, to provide doses of 2 and 10 mg/kg when given subcutaneously (SC) in a 5 mL/kg dosing volume. Control groups (n=5) received Vehicle (RNase-free PBS) SC at 5 mL/kg dosing volume.

Liver Processing for RT-qPCR

[0148] At Day 2 (48 hrs) and Day 7 (168 hrs) following siRNA compound or Vehicle injection (n=5), each treatment group was terminally sampled by cardiac puncture under isoflurane. Liver tissue was excised and snap frozen in liquid N.sub.2 Total RNA was extracted from homogenates of snap-frozen whole liver using GenElute™ Total RNA Purification Kit (RNB100-100RXN) Duplex RT-qPCR was performed using the ThermoFisher TaqMan Fast 1-Step Master Mix with TaqMan probes for GAPDH (VIC_PL), PCSK9 (FAM) and mTTR (FAM). Relative quantification (RQ) of PCSK9 was determined using the $\Delta\Delta CT$ method, where GAPDH was used as internal control and the expression changes of the target gene were normalized to the vehicle control.

Example 1

Testing 5' Versus 3' Positioning of Crook on the Sense Strand (SS) of Unmodified 'Inclisiran' Sequence in Serum Stability Assays

[0149] Following a 2 hr incubation in 10% FBS or 10% human serum, unmodified 'Inclisiran' with Crook positioned either at the 5' or 3' end of the SS, shows increased target mRNA (PCSK9) knockdown (KD) compared to the 'no crook' siRNA. However, superior KD is observed when crook is on the 5' end compared to 3' end of SS, following pre-treatment in human serum. This is demonstrated in FIG. 1, where 5' SS crook siRNA [striped bar] containing hairpin sequence GCGAAGC, maintains high levels of target KD (85%) in HepG2 cells following 2 hr treatment with 10% human serum comparable to that observed with modified Inclisiran (80%) [white bar]. Similar results can be shown when 'reversed' crook hairpin (CGAAGCG) is placed at the 5' end of the SS [spotted bar]. In contrast, 3' SS positioned Crook [hatched bar] shows -18.75% (loss of target KD) in HepG2 cells following pre-treatment in human serum; 65% KD (compared to 80% KD with no serum incubation). As expected, unmodified 'Inclisiran' with no Crook attached [grey bar] shows reduced levels of target KD after pre-treatment in either FBS or human serum: 50% and 60% KD, respectively, equating to -26.8% and -39% loss of KD.

Example 2

Testing 5' Positioning of Crook on the Sense Strand (SS) of Unmodified 'Inclisiran' Sequence in Serum Stability Assays with Increasing Concentrations of FBS

[0150] Following 2 hr incubations at 37 C in increasing concentrations of FBS, unmodified 'Inclisiran' sequence with Crook positioned at the 5' end of the SS [striped bar] shows sustained target mRNA (PCSK9) knockdown (70-80% KD) in all concentrations of FBS tested (10%, 20% and 50%), comparable to levels observed with modified Inclisiran (70-80% KD) [white bar]. Similarly, 'reversed' crook hairpin (CGAAGCG) on the 5' end of SS provides 65-75% KD with no loss of KD [spotted bar]. In contrast, the 'no crook' compound [grey bar] displays up to -85% loss of KD as only 20-50% target KD, is evident following serum treatment.

Example 3

Testing 5' Positioning of Crook on the Sense Strand (SS) of Unmodified 'Inclisiran' Sequence in Serum Stability Assays Over a 4-Hr Incubation Period

[0151] After a 4 hr incubation in either 10% FBS or 10% human serum, unmodified 'Inclisiran' with Crook positioned at the 5' end of the SS, shows sustained levels of approx. 75% target mRNA (PCSK9) knockdown (KD), and 65% KD, respectively [striped bar]. Similarly, there is no loss of KD evident for 'reversed' crook hairpin (CGAAGCG) on the 5' end of SS [spotted bar], comparable with modified 'Inclisiran', where approximately 70% KD is observed [white bar]. In contrast, the absence of Crook [grey bar] leads to substantially lower levels of KD following 4 hr

pre-treatment in 10% FBS (45% KD) or 10% human serum (35% KD), equating to a -36% and -50% loss in KD, respectively.

Example 4

Testing 5' Versus 3' Positioning of Crook on an Unmodified siRNA Sequence Targeting Pcsk9, in Serum Stability Assays (Sequence Termed PC8-18)

[0152] Following a 2 hr incubation in 10% FBS or 10% human serum, PC8-18 with Crook positioned at the 5' end of the sense strand (SS), shows superior levels of knockdown (KD) of target mRNA (PCSK9) compared to 3' positioned Crook on either the SS or AS. This is shown in FIG. 4, where there is sustained target KD (approx. 85%) for PC8-18 siRNA with 5'SS Crook: [horizontal striped bar & spots on black background bar] compared to 60-70% KD (equating to a loss of 30% KD compared to no serum treatment) seen with 3' SS positioned Crook [spots on white background bar & hatched bar]. Similarly, when Crook is placed on 3' AS, loss of KD is 6-16% resulting in 65-75% target KD [vertical striped bar]. When Crook is not present on PC8-18 siRNA, target KD is reduced to only 35% following 2 hr incubation in FBS equating to a substantial loss of KD (-63% compared to no serum treatment), and to only 25% KD in human serum (-77%). Similarly, uncrooked molecules that contain 3' dTdT overhangs, show loss of KD levels of -44% and -72% (compared to no serum treatment) following pre-treatment in FBS and human serum, respectively.

Example 5 Testing the In Vivo Silencing Effect of 5' Versus 3' Positioning of Crook on an Unmodified siRNA Compound Targeting PCSK9 (PC2 Sequence)

[0153] Groups of 5 mice for each treatment group were injected subcutaneously (SC) with either vehicle (PBS), compound A (no Crook), compound G (Crook on 5' end of sense strand (SS)), or compound H (Crook on 3' end of SS). Each compound was given at either 2 mg/kg or 10 mg/kg, and following sacrifice, levels of liver PCSK9 mRNA were measured at two time points (day 2 and day 7).

[0154] Compound G (5' SS Crook) results in 40% KD of PCSK9 mRNA in the liver after 48 hours at 2 and 10 mg/kg and 30% KD at 10 mg/kg after 7 days, compared to vehicle controls (FIG. 5A). Comparable liver target KD is seen after 48 hrs for compound H (3' SS Crook) approx. 50% KD at 2 mg/kg (30% KD at 10 mg/kg), with no significant KD observed at day 7 (FIG. 5A). Compound A which contains no Crook, shows noticeably less target KD, with no silencing following SC injection of 2 mg/kg dose at either 2 or 7 days. At the 10 mg/kg dose, compound A shows and <20% KD after 48 hrs, and 40% after 7 days (FIG. 5A).

Example 6

Testing Compounds a, G and H in Serum Stability Assays (HepG2 Cells)

[0155] Comparable results are shown for both 5' and 3' positioned Crook on the SS. Compound G (5' SS Crook) and compound H (3' SS Crook) maintains PCSK9 mRNA KD of >50% following a 2 hr incubation in either 10% FBS or human serum (compared to no serum treatment). In contrast, there is loss of target KD seen for compound A (no Crook), from 50% to only 20% KD following a 2 hr serum treatment; FIG. 5B.

[0156] When these siRNA compounds were further challenged in increasing serum concentrations (20% and 50%) over a 2 hr period, compound G (5'SS Crook) displayed superior performance over 3'SS positioned Crook (H) in human serum. This is shown in FIG. 5C, where a sustained level of target mRNA KD (approx. 50%) is evident only in compound G [spotted bar] following 2 hrs incubation in 50% human serum. This equates to no loss of KD for G when compared to its 'no serum' treatment KD level. In contrast, compound H [grey bar] shows a complete loss in KD (0%) performing exactly as 'no crook' compound A [white bar] after 2 hrs in 50% human serum.

TABLE-US-00001 TABLE 1 Selection of Lp(a) candidate siRNA sequences to which crook is conjugated

siRNA ID (SEQ ID NO)	Sense sequence (5'-3')	Antisense sequence (5'-3')
LP1 (43)	GCCCCUUAUUGUUAUACG	CGUAUAACAAUAAGGGGC
LP2 (44)	LP10	

(3) GCCCCUUAUUGUUAUACGA UCGUAUAACAAUAGGGGC (SEQ ID NO: 44)
(SEQ ID NO: 3) LP3 (45) LP11 (4) GCCCCUUAUUGUUAUACGA
TCGUUAUACAAUAAGGGGC (SEQ ID NO: 45) (SEQ ID NO: 4) LP4 (46)
LP12 (5) CCCCUUAUUGUUAUACGA UCGUAUAACAAUAAGGGG (SEQ ID NO:
46) (SEQ ID NO: 5) LPS (47) LP13 (6) CCCUUAUUGUUAUACGA
UCGUUAUACAAUAAGGG (SEQ ID NO: 47) (SEQ ID NO: 6) LP6 (48) LP14
(7) CCCCUUAUUGUUAUACA UGUUAUACAAUAAGGGG (SEQ ID NO: 48) (SEQ
ID NO: 7) LP7 (49) LP15 (41) CGGUAAUGGACAGAGUUUAU
AUAACUCUGUCCAUAUACCG (SEQ ID NO: 49) (SEQ ID NO: 41) LP8 (42)
LP16 (35) ACAGCCCCUUAUUGUUAUACGA CGUAUAACAAUAAGGGGC (SEQ ID
NO: 42) (SEQ ID NO: 35)

TABLE-US-00002 TABLE 2 Selection of APOC III and DGAT 2 siRNA
sequences to which crook is conjugated

SEQ ID NO	Name	Sequence
50	APOC3_01	5'-ACGGGACAGUAUUCUCAGUNA
51	APOC3_02	5'-CCCAAUAAAGCUGGACAAGAA
52	APOC3_03	5'-CUGUAGGUUGCUUAAAAGGGA
53	APOC3_04	5'-CUGGAGCACCGUUAAGGACAA
54	APOC3_05	5'-UCCCAAUAAAGCUGGACAAGA
55	APOC3_06	5'-GCCCCUGUAGGUUGCUUAAA
56	APOC3_07	5'-CCCUGAAAGACUACUGGAGCA
57	APOC3_08	5'-UGC UUAAAAGGGACAGUAUUC
58	APOC3_09	5'-GACCUCAAUACCCCAAGUCCA
59	APOC3_10	5'-GAGCACCGUUAAGGACAAGUU
60	APOC3_01	5'-ACGGGACAGUAUUCUCAGUNA
61	APOC3_02	5'-CCCAAUAAAGCUGGACAAGAA
62	APOC3_03	5'-CUGUAGGUUGCUUAAAAGGGA
63	APOC3_04	5'-CUGGAGCACCGUUAAGGACAA
64	APOC3_05	5'-UCCCAAUAAAGCUGGACAAGA
65	APOC3_06	5'-GCCCCUGUAGGUUGCUUAAA
66	APOC3_07	5'-CCCUGAAAGACUACUGGAGCA
67	APOC3_08	5'-UGC UUAAAAGGGACAGUAUUC
68	APOC3_09	5'-GACCUCAAUACCCCAAGUCCA
69	APOC3_10	5'-GAGCACCGUUAAGGACAAGUU
70	APOC3_01	5'-UCACUGAGAAUACUGUCCCGU-3'
71	APOC3_02	5'-UUCUUGUCCAGCUUUUAUUGGG-3'
72	APOC3_03	5'-UCCCUUUUAAGCAACCUACAG-3'
73	APOC3_04	5'-UUGUCCUUAACGGUGCUCCAG-3'
74	APOC3_05	5'-UCUUGUCCAGCUUUUAUUGGGA-3'
75	APOC3_06	5'-UUUUAAGCAACCUACAGGGGC-3'
76	APOC3_07	5'-UGC UCCAGUAGUCUUCAGGG-3'
77	APOC3_08	5'-GAUACUGUCCCUUUUAAGCA-3'
78	APOC3_09	5'-UGGACUUGGGGUUAUUGAGGUC-3'
79	APOC3_10	5'-AACUUGUCCUUAACGGUGCUC-3'
80	APOC3_01	5'-UCACUGAGAAUACUGUCCCGU
81	APOC3_02	5'-UUCUUGUCCAGCUUUUAUUGGG
82	APOC3_03	5'-UCCCUUUUAAGCAACCUACAG
83	APOC3_04	5'-UUGUCCUUAACGGUGCUCCAG
84	APOC3_05	5'-UCUUGUCCAGCUUUUAUUGGGA
85	APOC3_06	5'-UUUUAAGCAACCUACAGGGGC
86	APOC3_07	5'-UGC UCCAGUAGUCUUCAGGG
87	APOC3_08	5'-GAUACUGUCCCUUUUAAGCA
88	APOC3_09	5'-UGGACUUGGGGUUAUUGAGGUC
89	APOC3_10	5'-AACUUGUCCUUAACGGUGCUC
90	DGAT2_01	5'-CUCUGUAAA UUGGAAGUGUC
91	DGAT2_02	5'-CACCAUGAGCUAGGUGGAGUA
92	DGAT2_03	5'-UUCCUGAAGUGACAAAGGAAA
93	DGAT2_04	5'-GACCACCAGGAACUAUAUCUU
94	DGAT2_05	5'-GUUCCAGAAAUACA UUGGUUU
95	DGAT2_06	5'-AACCGCAAGGGCUUUGUGAAA
96	DGAT2_07	5'-GAGCAAGAAGU UCCCAGGCAU
97	DGAT2_08	5'-CAGUAGUAGGCAUCUGGAAUG
98	DGAT2_09	5'-GUCAUGGGUGUCUGUGGGUUA
99	DGAT2_10	5'-GCUCUGUAAA UUGGAAGUGU
100	DGAT2_01	5'-CUCUGUAAA UUGGAAGUGUC
101	DGAT2_02	5'-CACCAUGAGCUAGGUGGAGUA
102	DGAT2_03	5'-UUCCUGAAGUGACAAAGGAAA
103	DGAT2_04	5'-GACCACCAGGAACUAUAUCUU
104	DGAT2_05	5'-GUUCCAGAAAUACA UUGGUUU
105	DGAT2_06	5'-AACCGCAAGGGCUUUGUGAAA
106		

DGAT2_07 5'-GAGCAAGAGUAGUAGGCAU 107 DGAT2_08 5'-
 CAGUAGUAGGCAUCUGGAAUG 108 DGAT2_09 5'-GUCAUGGGUGUCUGUGGGUUA 109
 DGAT2_10 5'-GCUCUGUAAAUUUGGAAGUGU 110 DGAT2_01 5'-
 GACACUUCCAAAUUUACAGAG-3' 111 DGAT2_02 5'-UACUCCACCUAGCUCAUGGUG-3'
 112 DGAT2_03 5'-UUUCCUUUGUCACUUCAGGAA-3' 113 DGAT2_04 5'-
 AAGAUAUAGUUCCUGGUGGUC-3' 114 DGAT2_05 5'-AAACCAAUGUAUUUCUGGAAC-
 3' 115 DGAT2_06 5'-UUUCACAAAGCCCUUGCGGUU-3' 116 DGAT2_07 5'-
 AUGCCUGGGAACUUCUUGCUC-3' 117 DGAT2_08 5'-CAUUCCAGAUGCCUACUACUG-3'
 118 DGAT2_09 5'-UAACCCACAGACACCCAUGAC-3' 119 DGAT2_10 5'-
 ACACUUCCAAAUUUACAGAGC-3' 120 DGAT2_01 5'-GACACUUCCAAAUUUACAGAG
 121 DGAT2_02 5'-UACUCCACCUAGCUCAUGGUG 122 DGAT2_03 5'-
 UUUCCUUUGUCACUUCAGGAA 123 DGAT2_04 5'-AAGAUAUAGUUCCUGGUGGUC 124
 DGAT2_05 5'-AAACCAAUGUAUUUCUGGAAC 125 DGAT2_06 5'-
 UUUCACAAAGCCCUUGCGGUU 126 DGAT2_07 5'-AUGCCUGGGAACUUCUUGCUC 127
 DGAT2_08 5'-CAUUCCAGAUGCCUACUACUG 128 DGAT2_09 5'-
 UAACCCACAGACACCCAUGAC 129 DGAT2_10 5'-ACACUUCCAAAUUUACAGAGC
 TABLE-US-00003 TABLE 3 Selection of DGAT2 siRNA sequences (SEQ ID
 NOs 131-170), PCSK9 (SEQ ID NO: 171-210 and ApoCIII (SEQ ID NO:
 211 to 250) SEQ ID NO Sequence 131 GACCACCAGGAACUAUAUCUU sense
 sequence 132 GUUCCAGAAAUACAUUGGUUU sense sequence 133
 AACCGCAAGGGCUUUGUGAAA sense sequence 134 GAGCAAGAAGUUCCCAGGCAU
 sense sequence 135 CUUUGGAGAGAAUGAAGUGUA sense sequence 136
 CUUCGACAAGCACAAAGACCAA sense sequence 137 GCCGAUGGGUCCAGAAGAAGU
 sense sequence 138 CUUCACUUGGCUGGUGUUUGA sense sequence 139
 CUCCUUUGGAGAGAAUGAAGU sense sequence 140 UGCCAUCCUCAUGUACAUAUU
 sense sequence 141 CCGCAAGGGCUUUGUGAAACU sense sequence 142
 AGCAAGAAGUUCCCAGGCAUA sense sequence 143 AGUGUACAAGCAGGUGAUCUU
 sense sequence 144 UGCUGACCACCAGGAACUAUA sense sequence 145
 CCGAUGGGUCCAGAAGAAGUU sense sequence 146 UUUGGAGAGAAUGAAGUGUAC
 sense sequence 147 UGGCGCUACUUUCGAGACUAC sense sequence 148
 AAUGCCUGUGUUGAGGGAGUA sense sequence 149 AGUUCCAGAAAUACAUUGGUU
 sense sequence 150 CAGAAGUGAGCAAGAAGUUCC sense sequence 151
 AAGAUAUAGUUCCUGGUGGUC antisense sequence 152
 AAACCAAUGUAUUUCUGGAAC antisense sequence 153
 UUUCACAAAGCCCUUGCGGUU antisense sequence 154
 AUGCCUGGGAACUUCUUGCUC antisense sequence 155
 UACACUUCAUUCUCUCCAAAG antisense sequence 156
 UUGGUCUUGUGCUUGUCGAAG antisense sequence 157
 ACUUCUUCUGGACCCAUCGGC antisense sequence 158
 UCAAACACCAGCCAAGUGAAG antisense sequence 159
 ACUUCAUUCUCUCCAAAGGAG antisense sequence 160
 AAUAUGUACAUGAGGAUGGCA antisense sequence 161
 AGUUUCACAAAGCCCUUGCGG antisense sequence 162
 UAUGCCUGGGAACUUCUUGCUC antisense sequence 163
 AAGAUCACCUGCUUGUACACU antisense sequence 164
 UAUAGUUCCUGGUGGUCAGCA antisense sequence 165
 AACUUCUUCUGGACCCAUCGG antisense sequence 166
 GUACACUUCAUUCUCUCCAAA antisense sequence 167
 GUAGUCUCGAAAGUAGCGCCA antisense sequence 168
 UACUCCCUCAACACAGGCAUU antisense sequence 169

AACCAAUUUCUGAACU antisense sequence 170
 GGAACUUCUUGCUCACUUCUG antisense sequence 171
 CCUCAUAGGCCUGGAGUUUAU sense sequence 172 AGGCCUGGAGUUUAUUCGGAA
 sense sequence 173 CCCUCAUAGGCCUGGAGUUUA sense sequence 174
 AGGUCUGGAAUGCAAAGUCAA sense sequence 175 GGCCUGGAGUUUAUUCGGAAA
 sense sequence 176 CAGGUCUGGAAUGCAAAGUCA sense sequence 177
 CCUCACCAAGAUCUGCAUGU sense sequence 178 ACCUCAUAGGCCUGGAGUUU
 sense sequence 179 CACCAGCAUACAGAGUGACCA sense sequence 180
 AUCUCCUAGACACCAGCAUAC sense sequence 181 UCCUAGACACCAGCAUACAGA
 sense sequence 182 CUGGAGUUUAUUCGGAAAAGC sense sequence 183
 GCCUGGAGUUUAUUCGGAAAA sense sequence 184 GAGGCAGAGACUGAUCCACUU
 sense sequence 185 UAGGCCUGGAGUUUAUUCGGA sense sequence 186
 CACUUCUCUGCCAAAGAUGUC sense sequence 187 AUGCAAAGUCAAGGAGCAUGG
 sense sequence 188 GGUCAUGGUCACCGACUUCGA sense sequence 189
 GGCAGCUGUUUUGCAGGACUG sense sequence 190 GGGCAGGUUGGCAGCUGUUUU
 sense sequence 191 AUAAACUCCAGGCCUAUGAGG antisense sequence 192
 UUCCGAAUAAACUCCAGGCCU antisense sequence 193
 UAAACUCCAGGCCUAUGAGGG antisense sequence 194
 UUGACUUUGCAUUCCAGACCU antisense sequence 195
 UUUCCGAAUAAACUCCAGGCC antisense sequence 196
 UGACUUUGCAUUCCAGACCUG antisense sequence 197
 ACAUGCAGGAUCUUGGUGAGG antisense sequence 198
 AAACUCCAGGCCUAUGAGGGU antisense sequence 199
 UGGUCACUCUGUAUGCUGGUG antisense sequence 200
 GUAUGCUGGUGUCUAGGAGAU antisense sequence 201
 UCUGUAUGCUGGUGUCUAGGA antisense sequence 202
 GCUUUUCCGAAUAAACUCCAG antisense sequence 203
 UUUUCCGAAUAAACUCCAGGC antisense sequence 204
 AAGUGGAUCAGUCUCUGCCUC antisense sequence 205
 UCCGAAUAAACUCCAGGCCUA antisense sequence 206
 GACAUCUUUGGCAGAGAAGUG antisense sequence 207
 CCAUGCUCUUGACUUUGCAU antisense sequence 208
 UCGAAGUCGGUGACCAUGACC antisense sequence 209
 CAGUCCUGCAAAACAGCUGCC antisense sequence 210 AAAACAGCUGCCAACCUGCCC
 antisense sequence 211 CUGGAGCACCGUUAAGGACAA sense sequence 212
 CCCUGAAAGACUACUGGAGCA sense sequence 213 GAGCACCGUUAAGGACAAGUU
 sense sequence 214 ACUGGAGCACCGUUAAGGACA sense sequence 215
 CCUGAAAGACUACUGGAGCAC sense sequence 216 AAGACUACUGGAGCACCGUUA
 sense sequence 217 CAGUUCCCUGAAAGACUACUG sense sequence 218
 GGUGACCGAUGGCUUCAGUUC sense sequence 219 GGGUGACCGAUGGCUUCAGUU
 sense sequence 220 ACUACUGGAGCACCGUUAAGG sense sequence 221
 GACUACUGGAGCACCGUUAAG sense sequence 222 UUCAGUUCCCUGAAAGACUAC
 sense sequence 223 GUUCCCUGAAAGACUACUGGA sense sequence 224
 UGGAGCACCGUUAAGGACAAG sense sequence 225 CGCCACCAAGACCGCCAAGGA
 sense sequence 226 GGGCUGGGUGACCGAUGGCUU sense sequence 227
 GCCACCAAGACCGCCAAGGAU sense sequence 228 AGACUACUGGAGCACCGUUA
 sense sequence 229 CCACCAAGACCGCCAAGGAUG sense sequence 230
 UCCCUGAAAGACUACUGGAGC sense sequence 231 UUGUCCUUAACGGUGCUCCAG
 antisense sequence 232 UGCUCCAGUAGUCUUUCAGGG antisense sequence 233
 AACUUGUCCUUAACGGUGCUC antisense sequence 234

UGUCCUUAAGGUCUCCAGU	antisense	sequence 235
GUGCUCCAGUAGUCUUUCAGG	antisense	sequence 236
UAACGGUGCUCCAGUAGUCUU	antisense	sequence 237
CAGUAGUCUUUCAGGGAACUG	antisense	sequence 238
GAACUGAAGCCAUCGGUCACC	antisense	sequence 239
AACUGAAGCCAUCGGUCACCC	antisense	sequence 240
CCUUAACGGUGCUCCAGUAGU	antisense	sequence 241
CUUAACGGUGCUCCAGUAGUC	antisense	sequence 242
GUAGUCUUUCAGGGAACUGAA	antisense	sequence 243
UCCAGUAGUCUUUCAGGGAAC	antisense	sequence 244
CUUGUCCUUAACGGUGCUCCA	antisense	sequence 245
UCCUUGGCGGUCUUGGUGGCG	antisense	sequence 246
AAGCCAUCGGUCACCCAGCCC	antisense	sequence 247
AUCCUUGGCGGUCUUGGUGGC	antisense	sequence 248
UUAACGGUGCUCCAGUAGUCU	antisense	sequence 249
CAUCCUUGGCGGUCUUGGUGG	antisense	sequence 250
GCUCCAGUAGUCUUUCAGGGA	antisense	sequence
TABLE-US-00004 TABLE 4 siRNAs pairs used in silencing of APOC3 and DGAT 2 gene expression in HEPG2 cells in vitro		
Name	Sense	Antisense
APOC3_5'-5'-01	ACGGGACAGUAUUCUCAGUNA	Ucaccctcatccc UCACUGAGAAUACUGUCC gcgaagc-3'
(SEQ ID NO 401)	CGU-3' (SEQ ID NO 70)	
APOC3_5'-5'-02	CCCAAUAAAGCUGGACAAGAA	Ucaccctcatccc UUCUUGUCCAGCUUUUAUU gcgaagc-3'
(SEQ ID NO 402)	GGG-3' (SEQ ID NO 71)	
APOC3_5'-5'-03	CUGUAGGUUGCUUAAAAGGGA	Ucaccctcatccc UCCCUUUUAAGCAACCUA gcgaagc-3'
(SEQ ID NO 403)	CAG-3' (SEQ ID NO 72)	
APOC3_5'-5'-04	CUGGAGCACCGUUAAGGACA	Ucaccctcatccc UUGUCCUUAACGGUGCU gcgaagc-3' (SEQ ID NO 404)
CCAG-3' (SEQ ID NO 73)		
APOC3_5'-5'-05	UCCCAAUAAAGCUGGACAAGA	Ucaccctcatccc UCUUGUCCAGCUUUUAUUG gcgaagc-3'
(SEQ ID NO 405)	GGA-3' (SEQ ID NO 74)	
APOC3_5'-5'-06	GCCCCUGUAGGUUGCUUAAA	Ucaccctcatccc UUUUAAGCAACCUACAGG gcgaagc-3'
(SEQ ID NO 406)	GGC-3' (SEQ ID NO 75)	
APOC3_5'-5'-07	CCCUGAAAGACUACUGGAGCA	Ucaccctcatccc UGCUCCAGUAGUCUUUCA gcgaagc-3'
(SEQ ID NO 407)	GGG-3' (SEQ ID NO 76)	
APOC3_5'-5'-08	UGC UUAAAAGGGACAGUAUUC	Ucaccctcatccc GAAUACUGUCCCUUUUAA gcgaagc-3'
(SEQ ID NO 408)	GCA-3' (SEQ ID NO 77)	
APOC3_5'-5'-09	GACCUCAAUACCCCAAGUCCA	Ucaccctcatccc UGGACUUGGGGUAUUGA gcgaagc-3'
(SEQ ID NO 409)	GGUC-3' (SEQ ID NO 78)	
APOC3_5'-5'-10	GAGCACCGUUAAGGACAAGU	Ucaccctcatccc AACUUGUCCUUAACGGUG gcgaagc-3'
(SEQ ID NO 410)	CUC-3' (SEQ ID NO 79)	
DGAT2_5'-5'-01	CUCUGUAAAUUUGGAAGUGUC	Ucaccctcatccc GACACUCCAAAUUUACA gcgaagc-3'
(SEQ ID NO 411)	GAG-3' (SEQ ID NO 110)	
DGAT2_5'-5'-02	CACCAUGAGCUAGGUGGAGUA	Ucaccctcatccc UACUCCACCUAGCUCAUG gcgaagc-3'
(SEQ ID NO 412)	GUG-3' (SEQ ID NO 111)	
DGAT2_5'-5'-03	UUCCUGAAGUGACAAAGGAAA	Ucaccctcatccc UUUCCUUGUCACUUCAG gcgaagc-3'
(SEQ ID NO 413)	GAA-3' (SEQ ID NO 112)	
DGAT2_5'-5'-04	GACCACCAGGAACUAUAUCU	Ucaccctcatccc AAGAUUAUAGUCCUGGUG gcgaagc-3'
(SEQ ID NO 414)	GUC-3' (SEQ ID NO 113)	
DGAT2_5'-5'-05	GUUCCAGAAAUAACA UUGGUU	Ucaccctcatccc AAACCAAUGUAUUUCUGG gcgaagc-3'
(SEQ ID NO 415)	AAC-3' (SEQ ID NO 114)	
DGAT2_5'-5'-06	AACCGCAAGGGCUUUGUGAAA	Ucaccctcatccc UUUCACAAAGCCCUUGCG gcgaagc-3'

(SEQ ID NO 416) GUU-3'(SEQ ID NO 115) DGAT2_5'-5'-07
GAGCAAGAAGUUCCCAGGCAUtcacctcatccc AUGCCUGGGAACUUCUU gcgaagc-3'(SEQ ID NO 417) GCUC-3'(SEQ ID NO 116) DGAT2_5'-5'-08
CAGUAGUAGGCAUCUGGAAUGtcacctcatccc CAUUCCAGAUGCCUACUA gcgaagc-3'(SEQ ID NO 418) CUG-3'(SEQ ID NO 117) DGAT2_5'-5'-09
GUCAUGGGUGUCUGUGGGUUAtcacctcatcc UAACCCACAGACACCCAU cgcaagc-3'(SEQ ID NO 419) GAC-3'(SEQ ID NO 118) DGAT2_5'-5'-10
GCUCUGUAAAUUUGGAAGUGUtcacctcatccc ACACUCCAAAUUUACAG gcgaagc-3'(SEQ ID NO 420) AGC-3'(SEQ ID NO 119)

TABLE-US-00005 TABLE 5 Crook structures tested in the serum stability assay for 5' crook siRNAs 14b to siRNA15-5'CR consist of unmodified 'inclisiran' sequence (C = crook; CR = reversed hairpin Crook). siRNAs 35-44 consist of PC8 sequence siRNAs A, G and H consist of PC2 sequence Oligo name Sequence siRNA14m Sense: 5' Cm*Um*Am Gm Am Cm Cf Um Gf Um t Um Um Gm Cm Um Um Um 'Inclisiran' Um Gm Um 3' (SEQ ID NO 494) Antisense: 5' Am*Cf*Am Af Af Af Gm Cf Am Af Am Af Cm Af Gm Gf Um Cf Um Am Gm* Am* Am 3' (SEQ ID NO 495) siRNA14b Sense (5'-3'): CUAGACCUGULUUGCUUUUGU (SEQ ID NO 389) Antisense (5'-3'): ACAAAGCAAACAGGUCUAGAA (SEQ ID NO 390) siRNA15b Sense (5'-3'): CUAGACCUGUtUUGCUUUUGU**tcacctcatcccgcgaagc** (SEQ ID NO 496) Antisense (5'-3'): ACAAAGCAAACAGGUCUAGAA (SEQ ID NO 390) siRNA15-5'C Sense (5'-3'): **cgaagcgcctactccact**CUAGACCUGUUUGCUUUUGU (SEQ ID NO 497) Antisense (5'-3'): ACAAAGCAAACAGGUCUAGAA(SEQ ID NO 390) siRNA15-5'CR Sense (5'-3'): **gcgaagcccctactccact**CUAGACCUGUTUUGCUUUUGU (SEQ ID NO 498) Antisense (5'-3'): ACAAAGCAAACAGGUCUAGAA(SEQ ID NO 390) siRNA35 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA(SEQ ID NO 278 and 262 and 176) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG(SEQ ID NO 272, 196, 334) siRNA36 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdT (SEQ ID NO 421) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT (SEQ ID NO 422) siRNA37 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (Seq ID NO 423) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334) siRNA 38 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 278 and 262 and 176) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (SEQ ID NO 424) siRNA39 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (Seq ID NO 423) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT (SEQ ID NO 422) siRNA40 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCAdTdT SEQ ID NO 421) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdCdAdCdCdTdCdAdTdCdCdCdGdCdGdAdAd GdC (SEQ ID NO 424) siRNA41 Sense (5'-3'): dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTCAGGUCUGGAAUGCAAAG UCA (SEQ ID NO 425) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334) siRNA42 Sense (5'-3'): dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTCAGGUCUGGAAUGCAAAG UCA (SEQ ID NO 425) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUGdTdT (SEQ ID NO 422) siRNA43 Sense (5'-3'): dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTCAGGUCUGGAAUGCAAAG

UCAdTdT (SEQ ID NO 426) Antisense (5'-3'): UGACUUUGCAUUCCAGACCUG (SEQ ID NO 272, 196, 334) siRNA44 Sense (5'-3'): CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 278 and 262 and 176) Antisense (5'-3'):

dCdGdAdAdGdCdGdCdCdCdTdAdCdTdCdCdAdCdTUGACUUUGCAUUCCAGAC CUG (SEQ ID NO 427) siRNA-A 5'-AGGCCUGGAGUUUAUUCGGAA GalNAc-3' (SEQ ID NO 172 and 256) 3'-ttUCCGGACCUCAAAUAAGCCUU-5' (SEQ ID NO 252 and 254) siRNA-G 5'-cgaagcgcctactccactA*G*GCCUGGAGUUUAUUCGGAA GalNAc-3' (SEQ ID NO 428) 3'-t*t*UCCGGACCUCAAAUAAGCC*U*U-5' (SEQ ID NO: 527) siRNA-H 5'-AGGCCUGGAGUUUAUUCGGAAtcacctcatcccgcgaagc-3' (SEQ ID NO 253) 3'-GalNAc UCCGGACCUCAAAUAAGCCUU-5' (SEQ ID NO 429) Legend: **c, g, a, t or dT, dG, dA, dC**: DNA bases A, G, C, U: RNA bases f: 2'-deoxy-2'-fluoro m: 2'-O-methyl *internucleotide linkage phosphorothioate (PS) GalNAc

Example 7 (Inclisiran, PCSK9 Sequence)

[0157] When crook was attached at the 5' end of the sense strand (siRNA15-5'C), the siRNA sequence maintained a full KD activity against the target PCSK9 comparable to the chemically modified version (siRNA4m) after 2-hour incubation in 10% FBS or human serum. Crook at the 3' end of the sense strand (siRNA5b) showed partial protection in HS. siRNA with short crook (harpin part only) at the 3' (siRNA15s7) and 5' end (Inc 03), as well as the stem 12-nt part only at the 5' end (INC_02), all showed significant loss of KD compared to the full 19-nt crook when transfected in HepG2 at 25 nM (Table 6 and 7).

TABLE-US-00006 TABLE 6 Sequence KD in no KD after % KD loss % KD loss name serum 10% FBS 10% HS in FBS in HS siRNA14m 78.7 79.6 79.9 0.0 0.0 siRNA14b 82.0 51.4 59.1 37.3 28.0 siRNA15b 80.1 80.6 65.9 0.0 17.6 siRNA15-5'C 86.5 87.6 84.7 0.0 2.0 siRNA15s7 80.9 44.8 51.4 44.6 36.5

TABLE-US-00007 TABLE 7 siRNA name KD in no serum KD in 10% HS % KD loss in HS siRNA14m 50.4 62.2 0.0 siRNA14b 50.1 28.7 42.8 siRNA15-5'C 50.6 58.6 0.0 INC_02 34.2 0.0 100.0 INC_03 44.7 0.0 100.0

TABLE-US-00008 TABLE 8 siRNA description siRNA name Description Sequence siRNA14m Fully chemically S (5'-3') Cm*Um*Am Gm Am Cm Cf Um Gf Um t Um Um Gm modified version Cm Um Um Um Um Gm Um (SEQ ID NO: 494) AS (5'-3') Am*Cf*Am Af Af Af Gm Cf Am Af Am Af Cm Af Gm Gf Um Cf Um Am Gm* Am* Am (SEQ ID NO: 495) siRNA14b No crook S (5'-3'): CUAGACCUGUtUUGCUUUUGU (SEQ ID NO: 389) AS (5'-3'): ACAAAGCAAAACAGGUCUAGAA (SEQ ID NO: 390) siRNA15b Crook on 3' S S (5'-3'): CUAGACCUGUtUUGCUUUUGU**tcacctcatcccgcgaagc** strand (SEQ ID NO: 496) AS (5'-3'): ACAAAGCAAAACAGGUCUAGAA (SEQ ID NO: 390) siRNA15-5'C Crook on 5' S S (5'-3'): **cgaagcgcctactccact**CUAGACCUGUtUUGCUUUUGU strand (SEQ ID NO: 497) AS (5'-3'): ACAAAGCAAAACAGGUCUAGAA (SEQ ID NO: 390) siRNA15s7 Harpin part only S (5'-3'): CUAGACCUGUtUUGCUUUUGU**gcgaagc** (SEQ ID NO on 3' S strand 430) AS (5'-3'): ACAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390) Inc_02 12-nt stem only, S (5'-3'): **ccctactccact**CUAGACCUGUtUUGCUUUUGU (SEQ ID no hairpin NO 431) AS (5'-3'): ACAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390) Inc 03 Hairpin on 5' S S (5'-3'): **cgaagcg**CUAGACCUGUtUUGCUUUUGU (SEQ ID NO strand 432) AS (5'-3'): ACAAAGCAAAACAGGUCUAGAA (SEQ ID NO 390)

Example 8 (P8-18 PCSK9 Sequence)

[0158] When Crook was attached at the 5' end of the sense strand (PC8_05), the siRNA sequence maintained a full KD activity against the target PCSK9 after 2-hour incubation in 80% HS which

was comparable to the level of KD observed with no serum pre-incubation. Crook at the 3' end of the sense strand (PC8_01) gave substantially reduced protection in HS showing 72% percentage loss of KD compared to no serum pre-incubation. siRNA with short crook (harpin part only) at the 3' (P8_03) and 5' end (PC8_11), as well as the stem 12-nt part only at the 5' end (PC8_10), all showed significant loss of KD compared to the full 19-nt crook when transfected in HepG2 at 25 nM (Table 9).

TABLE-US-00009 TABLE 9 siRNA name KD in no serum KD in 80% HS % KD loss in 80% HS
PC8_00 72.9 0.0 100.0 PC8_01 51.6 14.5 72.0 PC8_03 50.0 1.4 97.3 PC8_05 63.3 60.4 4.6
PC8_10 54.0 25.8 52.2 PC8_11 72.6 17.3 76.2

TABLE-US-00010 TABLE 10 siRNA name Description Sequence PC8_00 No crook S (5'-3'): CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO 262, 278, 176) AS (5'-3'): UGACUUUGCAUUC CAGACCUG (SEQ ID NO 272, 196, 334) PC8_01 Crook on 3' S S (5'-3'): CAGGUCUGGAAUGCAAAGUCA**tcacctcatcccgcgaagc** (SEQ strand ID NO 433) AS (5'-3'): UGACUUUGCAUUC CAGACCUG (SEQ ID NO 272, 196, 334) PC8_03 Crook hairpin S (5'-3'): CAGGUCUGGAAUGCAAAGUCA**gcgaagc** (SEQ ID NO 434) on 3' S strand AS (5'-3'): UGACUUUGCAUUC CAGACCUG (SEQ ID NO 272, 196, 334) PC8_05 Crook on 5' S S (5'-3'): **cgaagcgcctactccact**CAGGUCUGGAAUGCAAAGUCA (SEQ strand ID NO 435) AS (5'-3'): UGACUUUGCAUUC CAGACCUG (SEQ ID NO 272, 196, 334) PC8_10 12-nt stem S (5'-3'): **ccctactccact**CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO only, no 436) hairpin AS (5'-3'): UGACUUUGCAUUC CAGACCUG (SEQ ID NO 272, 196, 334) SIRNA_A No crook S (5'- 3') AGGCCUGGAGUUUAUUCGGAA GalNAc (SEQ ID NO 172, 256) AS (3'-5') ttUCCGGACCUC AAAUAAGCCUU (SEQ ID NO 252, 254) SIRNA_G Crook on 5' S (5'-3') S strand **cgaagcgcctactccact**A*G*GCCUGGAGUUUAUUCGGAA GalNAc (SEQ ID NO 437) AS (3'-5') t*t*UCCGGACCUC AAAUAAGCC*U*U (SEQ ID NO: 527) SIRNA_H Crook on 3' S (5'-3) S strand A*G*GCCUGGAGUUUAUUCGGAA**tcacctcatcccgcgaagc** (SEQ ID NO 438) AS (3'-5') GalNAc t*t*UCCGGACCUC AAAUAAGCC*U*U (SEQ ID NO: 528) PC8_11 Hairpin on 5' S (5'-3'): **cgaagcg**CAGGUCUGGAAUGCAAAGUCA (SEQ ID NO: 526) S strand AS (5'-3'): UGACUUUGCAUUC CAGACCUG (SEQ ID NO 272, 196, 334)

Example 9 (Compound G—PCSK9 Sequence)

[0159] When Crook was attached at the 5' end of the sense strand (siRNA_G), the siRNA sequence maintained a full KD activity against PCSK9 after 8-hour incubation in 80% HS comparable to the level of KD observed with no serum pre-incubation (Table 11). In contrast, siRNA_A (no crook) or siRNA_H (crook at the 3' end of the sense strand) showed no protection in 80% HS and a loss of % KD of 70.8% and 100% respectively when transfected in HepG2 at 25 nM. In a free-uptake assay, siRNA_G showed better KD levels compared to siRNA_H in primary mouse hepatocytes cultured in 10% FBS and treated for 24, 48 and 72 hours with 100 nM of siRNA (Table 12).

TABLE-US-00011 TABLE 11 siRNA name KD in no serum KD in 80% HS % KD loss in HS
siRNA_A 53.2 15.5 70.8 siRNA_G 59.5 58.5 1.7 siRNA_H 57.6 0.0 100.0

TABLE-US-00012 TABLE 12 KD levels of PCSK9 following free-uptake of siRNA_A, siRNA_H and siRNA_G at 100 nM in primary mouse hepatocytes cultured in 10% FBS % KD % KD % KD
Time from treatment siRNA_A siRNA_H siRNA_G 24 hours 0.0 0.0 15.0 48 hours 5.8 17.5 50.6
72 hours 0.0 0.0 36.5

Example 10 (ApoB Sequences)

[0160] When a total of 11 siRNAs carrying a sequence against mouse ApoB were exposed to 20% and 50% human serum, and subsequently transfected at 25 nM into primary mouse hepatocytes, the siRNA variants carrying 5' crook on the sense strand showed, overall, a better ability to induce

KOD of ApoB after exposure to serum compared to the siRNA variants carrying crook on the 3' end (Table 13).

TABLE-US-00013

TABLE 13	KD	KD	KD	KD	% KD	% Crook	% no	20%	50%	loss in	loss in																																																																																																														
siRNA Name	position	serum	HS	HS	20%	HS	50%	HS	TS3_1	5'SS	70.5	55.1	0.0	21.8	100.0	TS3_2																																																																																																									
3'SS	75.4	0.0	0.0	100.0	100.0	TS3_3	3'AS	73.1	0.0	0.0	100.0	100.0	TS4_1	5'SS	64.4	9.3	8.8	85.6																																																																																																							
86.3	TS4_2	3'SS	63.4	10.9	0.0	82.7	100.0	TS4_3	3'AS	67.9	0.0	0.0	100.0	100.0	ApoB_C10_1	5'SS	58.0	67.3	62.3	0.0	0.0	ApoB_C10_2	3'SS	52.4	0.0	38.2	100.0	27.0	ApoB_C10_3	3'AS	56.6	56.6																																																																																									
18.9	0.0	66.6	ApoB_C3_1	5'SS	48.1	50.9	37.6	0.0	21.7	ApoB_C3_2	3'SS	58.1	49.8	48.0	14.2	17.3	ApoB_C3_3	3'AS	54.9	41.2	0.0	25.0	100.0	ApoB_C2_1	5'SS	66.8	87.7	87.6	0.0	0.0	ApoB_C2_2	3'SS	70.9	87.1	86.8	0.0	0.0	ApoB_C2_3	3'AS	79.6	87.5	87.6	0.0	0.0	ApoB_DM2_1	5'SS	58.4	76.4																																																																									
67.4	0.0	0.0	ApoB_DM2_2	3'SS	64.0	65.4	4.3	0.0	93.3	ApoB_DM2_3	3'AS	67.6	55.3	5.0	18.1	92.5	ApoB_DM3_1	5'SS	72.0	56.4	0.0	21.7	100.0	ApoB_DM3_2	3'SS	75.4	0.0	0.0	100.0	100.0	ApoB_DM3_3	3'AS	82.7	0.0	0.0	100.0	100.0	ApoB_DM5_1	5'SS	51.6	63.7	73.4	0.0	0.0	ApoB_DM5_2	3'SS	59.5	0.0	0.0	100.0	100.0	ApoB_DM5_3	3'AS	55.7	74.9	78.4	0.0	0.0	ApoB_DM13_1	5'SS	84.8	87.2	49.1	0.0	42.1	ApoB_DM13_2	3'SS	88.6	50.3	45.6	43.3	48.6	ApoB_DM13_3	3'AS	87.0	41.3	23.7	52.6	72.7	ApoB_DM18_1	5'SS	81.8	70.8	50.9	13.4	37.8	ApoB_DM18_2	3'SS	83.4	80.0	36.2	4.1	56.6	ApoB_DM18_3	3'AS	86.8	32.4	7.8	62.6	91.0	ApoB_DM19_1	5'SS	60.0	1.7	0.0	97.1	100.0	ApoB_DM19_2	3'SS	68.6	0.0	0.0	100.0	100.0	ApoB_DM19_3	3'AS	74.4	0.0	0.0	100.0	100.0

TABLE-US-00014 ApoB sequences Table 14 siRNA name Description Sequence TS3_1 5'sense S (5'-3'): **cgaagcgccctactccact**UAGACUUCCUGAAUAAC*U*A (SEQ ID NO 439) AS (5'-3'): U*A*GUUAUUCAGGAAGUCUA*U*U (SEQ ID NO 440) TS3_2 3'sense S (5'-3'): U*A*GACUUCCUGAAUAACUA**tcacctcatcccgcgaagc** (SEQ ID NO 441) AS (5'-3'): U*A*GUUAUUCAGGAAGUCUA*U*U (SEQ ID NO 440) TS3_3 3'antisense S (5'-3'): U*A*GACUUCCUGAAUAAC*U*A (SEQ ID NO 442) AS (5'-3'): U*A*GUUAUUCAGGAAGUCUA*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 443) TS4_1 5'sense S (5'-3'): cgaagcgccctactccactUCAUCACACUGAAUACC*A*A (SEQ ID NO 444) AS (5'-3'): U*U*GGUAUUCAGUGUGAUGA*U*U (SEQ ID NO 445) TS4_2 3'sense S (5'-3'): U*C*AUCACACUGAAUACCA**Atcacctcatcccgcgaagc** (SEQ ID NO 446) AS (5'-3'): U*U*GGUAUUCAGUGUGAUGA*U*U (SEQ ID NO 445) TS4_3 3'antisense S (5'-3'): U*C*AUCACACUGAAUACC*A*A (SEQ ID NO 447) AS (5'-3'): U*U*GGUAUUCAGUGUGAUGA*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 448) ApoB_C10_1 5'sense S (5'-3'): **cgaagcgccctactccact**GUCAUCACACUGAAUACCA*A*U (SEQ ID NO 449) AS (5'-3'): A*U*UGGUAUUCAGUGUGAUGAC*U*U (SEQ ID NO 450) ApoB_C10_2 3'sense S (5'-3'): G*U*CAUCACACUGAAUACCA**Utcacctcatcccgcgaagc** (SEQ ID NO 451) AS (5'-3'): A*U*UGGUAUUCAGUGUGAUGAC*U*U (SEQ ID NO 450) ApoB_C10_3 3'antisense S (5'-3'): G*U*CAUCACACUGAAUACCA*A*U (SEQ ID NO 453) AS (5'-3'): A*U*UGGUAUUCAGUGUGAUGAC*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 452) ApoB_C3_1 5'sense S (5'-3'): **cgaagcgccctactccact**GGUGUAUGGCUUCAACCCU*G*A (SEQ ID NO 454) AS (5'-3'): U*C*AGGGUUGAAGCCAUAACACC*U*U (SEQ ID NO 455) ApoB_C3_2 3'sense S (5'-3'): G*G*UGUAUGGCUUCAACCCUGA**tcacctcatcccgcgaagc** (SEQ ID NO 456) AS (5'-3'): U*C*AGGGUUGAAGCCAUAACACC*U*U (SEQ ID NO 455) ApoB_C3_3 3'antisense S (5'-3'): G*G*UGUAUGGCUUCAACCCU*G*A (SEQ ID NO 457) AS (5'-3'): U*C*AGGGUUGAAGCCAUAACACC*U*U**tcacctcatcccgcgaagc** (SEQ ID NO 458) ApoB_C2_1 5'sense S (5'-3'): **cgaagcgccctactccact**CACCAACUUCUCCACGAG*U*C (SEQ ID NO 459) AS (5'-3'): G*A*CUCGUGGAAGAAGUUGGUG*U*U (SEQ ID NO 460) ApoB_C2_2 3'sense S (5'-3'): C*A*CCAACUUCUCCACGAGU**tcacctcatcccgcgaagc** (SEQ ID NO 461) AS

(5'-3'): G*A*CUUGGGAAGAAGUUGG*U*U (SEQ ID NO 460) ApoB_C2_3
3'antisense S (5'-3'): C*A*CCAACUUCUUCCACGAG*U*C (SEQ ID NO 462) AS
(5'-3'): G*A*CUUGGGAAGAAGUUGGUG*U*U **Utcacctcatccccggaagc** (SEQ ID NO
463) ApoB_DM2_1 5'sense S (5'-3'):
cgaagcgccctactccactAGGCAGAGCUAGUGGCA*A*A (SEQ ID NO 464) AS (5'-3'):
U*U*UGCCACUAGCUCUGCCU*U*U (SEQ ID NO 465) ApoB_DM2_2 3'sense S
(5'-3'): A*G*GCAGAGCUAGUGGCAAA**tcacctcatccccggaagc** (SEQ ID NO 466) AS
(5'-3'): U*U*UGCCACUAGCUCUGCCU*U*U (SEQ ID NO 465) ApoB_DM2_3
3'antisense S (5'-3'): A*G*GCAGAGCUAGUGGCA*A*A (SEQ ID NO 467) AS
(5'-3'): U*U*UGCCACUAGCUCUGCCU*U*U **Utcacctcatccccggaagc** (SEQ ID NO 468)
ApoB_DM3_1 5'sense S (5'-3'): **cgaagcgccctactccact**GAGCAAUCUCUUCAAU*A*A
(SEQ ID NO 469) AS (5'-3'): U*U*AUUGAAGAGAUUUGCUC*U*U (SEQ ID NO
470) ApoB_DM3_2 3'sense S (5'-3'): G*A*GCAAUCUCUUCAAUAA**tcacctcatccccggaagc**
(SEQ ID NO 471) AS (5'-3'): U*U*AUUGAAGAGAUUUGCUC*U*U (SEQ ID NO
470) ApoB_DM3_3 3'antisense S (5'-3'): G*A*GCAAUCUCUUCAAU*A*A (SEQ
ID NO 472) AS (5'-3'): U*U*AUUGAAGAGAUUUGCUC*U*U **Utcacctcatccccggaagc**
(SEQ ID NO 473) ApoB_DM5_1 5'sense S (5'-3'):
cgaagcgccctactccactCCACAAAUGUCUACAGC*A*A (SEQ ID NO 474) AS
(5'-3'): U*U*GCUGUAGACAUUUGUGG*U*U (SEQ ID NO 475) ApoB_DM5_2 3'sense
S (5'-3'): C*C*ACAAAUGUCUACAGCAAT**tcacctcatccccggaagc** (SEQ ID NO 476) AS
(5'-3'): U*U*GCUGUAGACAUUUGUGG*U*U (SEQ ID NO 475) ApoB_DM5_3
3'antisense S (5'-3'): C*C*ACAAAUGUCUACAGC*A*A (SEQ ID NO 477) AS
(5'-3'): U*U*GCUGUAGACAUUUGUGG*U*U **Utcacctcatccccggaagc** (SEQ ID NO 478)
ApoB_DM13_1 5'sense S (5'-3'): **cgaagcgccctactccact**GAAACAGGCUUGAAAGA*A*U
(SEQ ID NO 479) AS (5'-3'): A*U*UCUUUCAAGCCUGUUUC*U*U (SEQ ID NO
480) ApoB_DM13_2 3'sense S (5'-3'):
G*A*AACAGGCUUGAAAGAAU**tcacctcatccccggaagc** (SEQ ID NO 481) AS (5'-3'):
A*U*UCUUUCAAGCCUGUUUC*U*U (SEQ ID NO 480) ApoB_DM13_3 3'antisense S
(5'-3'): G*A*AACAGGCUUGAAAGAA*U (SEQ ID NO 482) AS (5'-3'):
A*U*UCUUUCAAGCCUGUUUC*U*U **Utcacctcatccccggaagc** (SEQ ID NO 483)
ApoB_DM18_1 5'sense S (5'-3'): **cgaagcgccctactccact**GAGAGAAAUCGAAGAGG*A*A
(SEQ ID NO 484) AS (5'-3'): U*U*CCUCUUCGAUUUCUCUC*U*U (SEQ ID NO
485) ApoB_DM18_2 3'sense S (5'-3'):
G*A*GAGAAAUCGAAGAGGAAT**tcacctcatccccggaagc** (SEQ ID NO 486) AS (5'-3'):
U*U*CCUCUUCGAUUUCUCUC*U*U (SEQ ID NO 485) ApoB_DM18_3 3'antisense S
(5'-3'): G*A*GAGAAAUCGAAGAGG*A*A (SEQ ID NO 487) AS (5'-3'):
U*U*CCUCUUCGAUUUCUCUC*U*U **Utcacctcatccccggaagc** (SEQ ID NO 488)
ApoB_DM19_1 5'sense S (5'-3'): **cgaagcgccctactccact**AGUUAUAGUCCGUGAGC*U*A
(SEQ ID NO 489) AS (5'-3'): U*A*GCUCACGGACUAUAACU*U*U (SEQ ID NO
490) ApoB_DM19_2 3'sense S (5'-3'):
A*G*UUAUAGUCCGUGAGCUAT**tcacctcatccccggaagc** (SEQ ID NO 491) AS (5'-3'):
U*A*GCUCACGGACUAUAACU*U*U (SEQ ID NO 490) ApoB_DM19_3 3'antisense S
(5'-3'): A*G*UUAUAGUCCGUGAGC*U*A (SEQ ID NO 492) AS (5'-3'):
U*A*GCUCACGGACUAUAACU*U*U **Utcacctcatccccggaagc** (SEQ ID NO 493)
TABLE-US-00015 TABLE 15 SEQ ID NO 3 auaacucugu ccauuaccg 8 ucguauaaca
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agaaugugcc ucgauaacu 13 auaacucugu ccaucacca 14 auaacucugu ccaucaccu 15 uaacucuguc
cauuaccu 16 augugccuug auaacucug 17 aguuggugcu gcuucagaa 18 aaauaggggc ugccacagg
19 uaacucuguc caucaccau 20 augagccucg auaacucug 21 aaugagccuc gauaacucu 22
aaugcuucca ggacauuuc 23 acaguggugg agaaugugc 24 guauggccu cgauaacuc 25

ucgauaacuc ugucacua 26 Ugucacugga cauuguguc 27 cugggaucca ugguguaac 28
 agaugaccaa gcuuggcag 29 uuaacucugu ccauuaccg 30 uuaacucugu ccauuacc 31
 uuaacucugu ccauuaccu 32 auaacucugu ccauuacc 33 auaacucugu ccauuaccu 34 uuaacucugu
 ccauuacca 36 ucguauaaca auaaggggc 37 tcguauaaca auaaggggc 38 ucguauaaca auaagggg 39
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 ggccuggagu uuauucggaa a 263 ccucaccaag auccugcaug u 264 caccagcaua cagagugacc a
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 aaaucgcuga uuuguguag 397 uaaggcuau aagagauact t 398 aaguaucucu ucauagccuu a 399
 aaaaaagcaa aacaggucua gaa 400 gcgaagcccc tactccact 427 cgaagcgccc tactccactu
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 A*U*UGGUAUUCAGUGUGAUGAC*U*U 502 GGUGUAUGGCUUCAACCCU*G*A 503
 G*G*UGUAUGGCUUCAACCCUGA 504 U*C*AGGGUUGAAGCCAUAACACC*U*U 505
 CACCAACUUCUUCCACGAG*U*C 506 C*A*CCAAUCUUCUCCACGAGUC 507
 G*A*CUCGUGGAAGAAGUUGGUG*U*U 508 AGGCAGAGCUAGUGGCA*A*A 509
 A*G*GCAGAGCUAGUGGCAAA 510 U*U*UGCCACUAGCUCUGCCU*U*U 511
 GAGCAAAUCUCUCAAU*A*A 512 G*A*GCAAAUCUCUCAAUAA 513
 U*U*AUUGAAGAGAUUUGCUC*U*U 514 CCACAAAUGUCUACAGC*A*A 515
 C*C*ACAAAUGUCUACAGCAA(516 U*U*GCUGUAGACAUUUGUGG*U*U 517
 GAAACAGGCUUGAAAGA*A*U 518 G*A*AACAGGCUUGAAAGAAU 519
 A*U*UCUUCAAGCCUGUUUC*U*U 520 GAGAGAAAUCGAAGAGG*A*A 521
 G*A*GAGAAAUCGAAGAGGAA 522 U*U*CCUCUUCGAUUUCUCUC*U*U 523
 AGUUAUAGUCCGUGAGC*U*A 524 A*G*UUAUAGUCCGUGAGCUA 525
 U*A*GCUCACGGACUAUAACU*U*U

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Claims

1. A nucleic acid molecule comprising: a first part that comprises a double stranded inhibitory ribonucleic acid (RNA) molecule comprising a sense strand and an antisense strand; and a second part that comprises a single stranded deoxyribonucleic acid (DNA) molecule, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule or wherein the 3' end of the single stranded DNA molecule is covalently linked to the 5' of the antisense strand of the double stranded inhibitory RNA molecule, characterized in that the double stranded inhibitory RNA comprises a sense nucleotide sequence that encodes a part of a cardiovascular gene target associated with cardiovascular disease, or a polymorphic sequence variant thereof, and wherein said single stranded DNA molecule comprises a nucleotide sequence that is adapted over at least part of its length to anneal by complementary base pairing to a part of said single stranded DNA to form a double stranded DNA structure comprising a stem and a loop domain, characterized in that said nucleic acid molecule comprises N-acetylgalactosamine and said double stranded inhibitory RNA consists of natural nucleotides.

2-3. (canceled)

4. The nucleic acid molecule according to claim 1, wherein said loop domain comprises the nucleotide sequence GCGAAGC.

5. The nucleic acid molecule according to claim 4 wherein said single stranded DNA molecule comprises the nucleotide sequence selected from the group: TABLE-US-00016 (SEQ ID NO 387 and 251) 5' TCACCTCATCCCGCGAAGC 3'; (SEQ ID NO 130) 5' CGAAGCGCCCTACTCCACT 3'; and (SEQ ID NO 400) 5' GCGAAGCCCCTACTCCACT 3'.

6. The nucleic acid molecule according to claim 1, wherein: said inhibitory RNA molecule comprises a two-nucleotide overhang comprising or consisting of at least one deoxythymidine dinucleotide (dTdT); said sense and/or said antisense strands comprises at least one internucleotide phosphorothioate linkages; and/or said nucleic acid molecule comprises a vinylphosphonate modification.

7-8. (canceled)

9. The nucleic acid molecule according to claim 1, wherein said double stranded inhibitory RNA molecule is between 17 and 29 nucleotides or 19 to 21 nucleotides in length.

10. The nucleic acid molecule according to claim 1, wherein said cardiovascular gene target is human Lipoprotein (a).

11. The nucleic acid molecule according to claim 10, wherein said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence selected from the group consisting of: SEQ ID NO: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33 and 34.

12. The nucleic acid molecule according to claim 10, wherein; said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 41 and a sense nucleotide sequence comprising SEQ ID NO: 49, wherein said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule; said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 4 and a sense nucleotide sequence comprising SEQ ID NO: 44, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule; or said double stranded inhibitory RNA molecule comprises an antisense nucleotide sequence comprising SEQ ID NO: 5 and a sense nucleotide sequence comprising SEQ ID NO: 46, wherein said single stranded DNA molecule is covalently linked to the 5' end of the antisense strand of the double stranded inhibitory RNA molecule.

13-14. (canceled)

15. The nucleic acid molecule according to claim 1, wherein said cardiovascular gene target is human Apolipoprotein C III (Apo C III).

16. The nucleic acid molecule according to claim 15, wherein; said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78 and 79; said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 211, 212, 213, 214, 215, 216, 217, 218, 219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, 230, 231, 232, 233, 234, 235, 236, 237, 238, 239, 240, 241, 242, 243, 244, 245, 246, 247, 248, 249 and 250; or said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 50, 51, 52, 53, 54, 55, 56, 57, 58, 80, 81, 82, 83, 84, 85, 86, 87, 88 and 89.

17-18. (canceled)

19. The nucleic acid molecule according to claim 1, wherein said cardiovascular gene target is human diglyceride acyltransferase 2 (DGAT2).

20. The nucleic acid molecule according to claim 19, wherein; said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 100,

101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118 and 119; said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 131, 132, 133, 134, 135, 136, 137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149, 150, 151, 152, 153, 154, 156, 157, 158, 159, 160, 161, 162, 163, 164, 165, 166, 167, 168, 169 and 170; or said nucleic acid comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 120, 121, 122, 123, 124, 125, 126, 127, 128 and 129.

21-22. (canceled)

23. The nucleic acid molecule according to claim 1, wherein said cardiovascular gene target is human PCSK9.

24. The nucleic acid molecule according to claim 23, wherein; said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 171, 172, 173, 174, 175, 176, 177, 178, 179, 180, 181, 182, 183, 184, 185, 186, 187, 189, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209 and 210; or said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 191, 192, 193, 194, 195, 196, 197, 198, 199, 200, 201, 202, 203, 204, 205, 206, 207, 208, 209 and 210.

25. (canceled)

26. The nucleic acid molecule according to claim 1, wherein said cardiovascular gene target is human Apolipoprotein B.

27. The nucleic acid molecule according to claim 26, wherein; said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 499, 500, 453, 502, 503, 457, 505, 506, 462, 508, 509, 467, 511, 512, 472, 514, 515, 477, 517 518, 482, 520, 521, 487, 523, 524 and 492; or said nucleic acid molecule comprises an RNA strand comprising a nucleotide sequence selected from the group consisting of: SEQ ID NO: 450, 501, 455, 504, 460, 507, 465, 510, 470, 513, 475, 516, 480, 519, 485, 522, 490 and 525.

28. (canceled)

29. The nucleic acid molecule according to claim 1, wherein said nucleic acid molecule is covalently linked to N-acetylgalactosamine.

30. A pharmaceutical composition comprising at least one nucleic acid molecule according to claim 1 and a pharmaceutically acceptable carrier or excipient.

31. The pharmaceutical composition according to claim 30 wherein said composition comprises at least one further therapeutic agent.

32. A method of treating a subject that has or is predisposed to hypercholesterolemia or diseases associated with hypercholesterolemia, comprising administering to the subject an effective dose of the nucleic acid molecule according to claim 1.

33. The nucleic acid molecule according to claim 1, wherein the 3' end of said single stranded DNA molecule is covalently linked to the 5' end of the sense strand of the double stranded inhibitory RNA molecule and said N-acetylgalactosamine is linked to the 3' end of said sense strand.
